

InsightEdge Newsletter - Customer Complaints

June 2026

Comprehensive Insights Across Locations Pune, Lucknow and Jamshedpur

Date: 23rd July 2026

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Note

For this week's newsletter, we have analyzed **Customer Complaint data** across all 3 cities between the timeframe **August 2024 to June 2026**.

This approach helps us provide actionable insights while maintaining uniformity in the analysis period.

Kindly find the **Month-wise analysis** for Customer Complaints data in the link below.

Link: <https://docs.google.com/spreadsheets/d/>

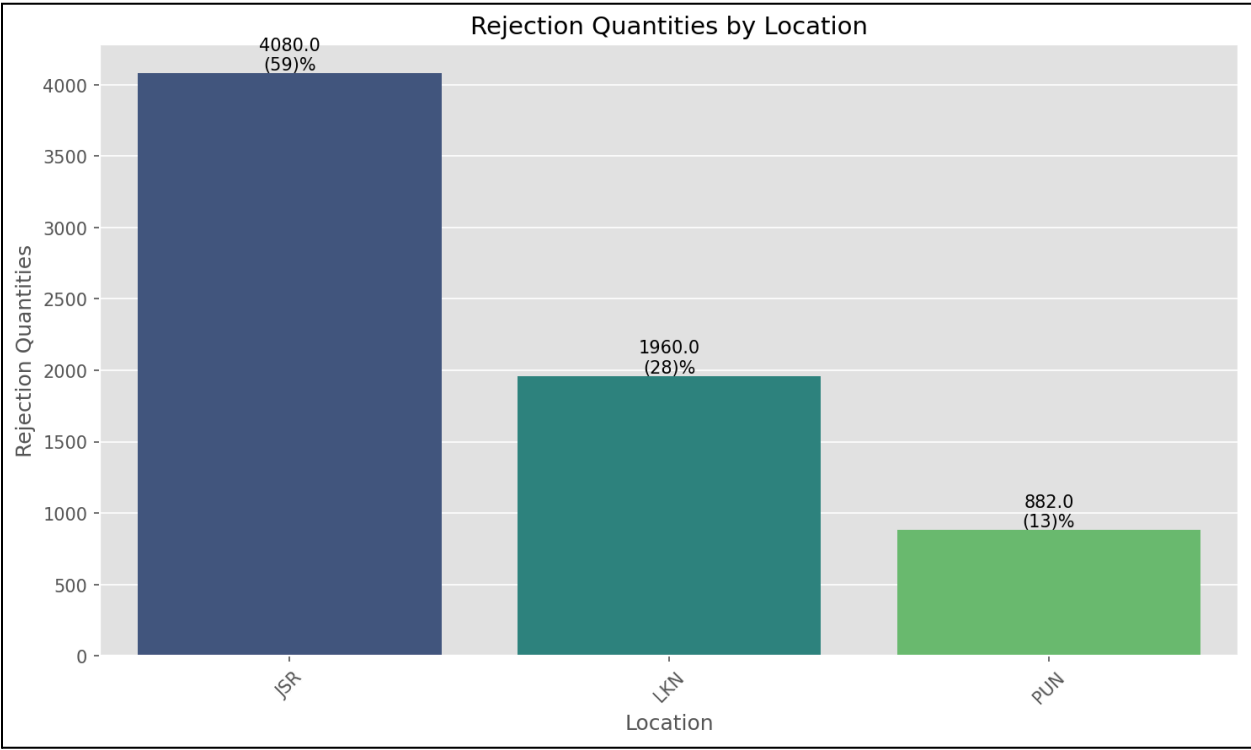
Please Note: The **highlighted texts** in the following queries can be replaced with specific user inputs to generate the required outputs/insights in the InsightEdge application.

Agenda 1:

Summary Until Generation Date

1. Location-Wise Summary

Query: Create a bar graph to represent the rejection quantities for locations: ['PUN', 'LKN', 'JSR'], spanning the months from August 2024 to May 2026. Label the x-axis with the names of the locations and the y-axis with the rejection quantities. Use distinct colors for each location to ensure clarity and differentiate the bars visually. Remove any black lines (error bars) and annotate each bar with its corresponding rejection quantity value and its percentage, rounded to real numbers for enhanced readability and presentation.

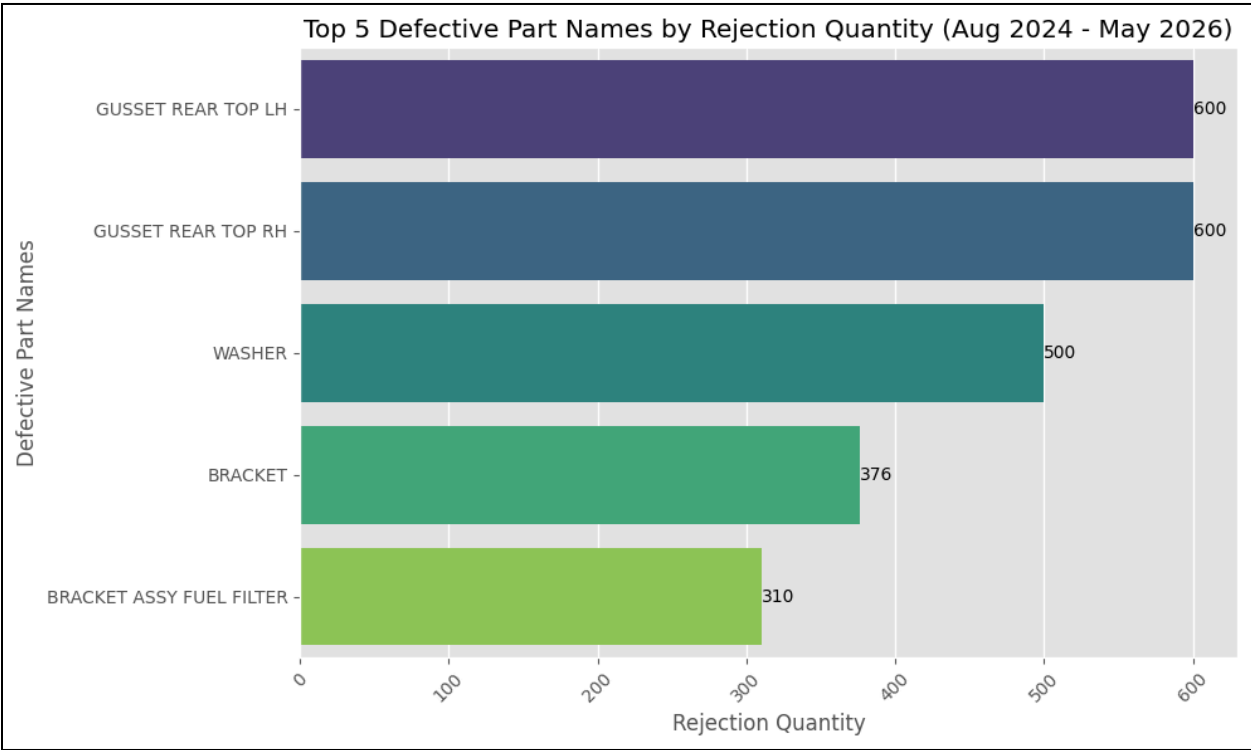


InsightEdge Observations:

- 1. **JSR Dominance:** JSR accounts for 59% of total rejections, significantly higher than other locations.
- 2. **LKN Contribution:** LKN represents 28% of rejections, showing a notable gap compared to JSR.
- 3. **PUN Minimal Impact:** PUN has the lowest contribution at 13%, indicating less issues compared to other locations.

2. Top Defective Parts

Query: Create a professional horizontal bar graph to visualize the **top 5** defective part names for all locations **['PUN', 'LKN', 'JSR']** from the month of **August 2024 to May 2026**. The x-axis should represent the rejection quantities, while the y-axis should display the defective part names. Each bar should correspond to a specific defective part name, with distinct colors or patterns, ensure clarity and easy differentiation. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability. Label the axes and ensure all the names are clearly visible and include a descriptive title to create a professional and visually appealing presentation.

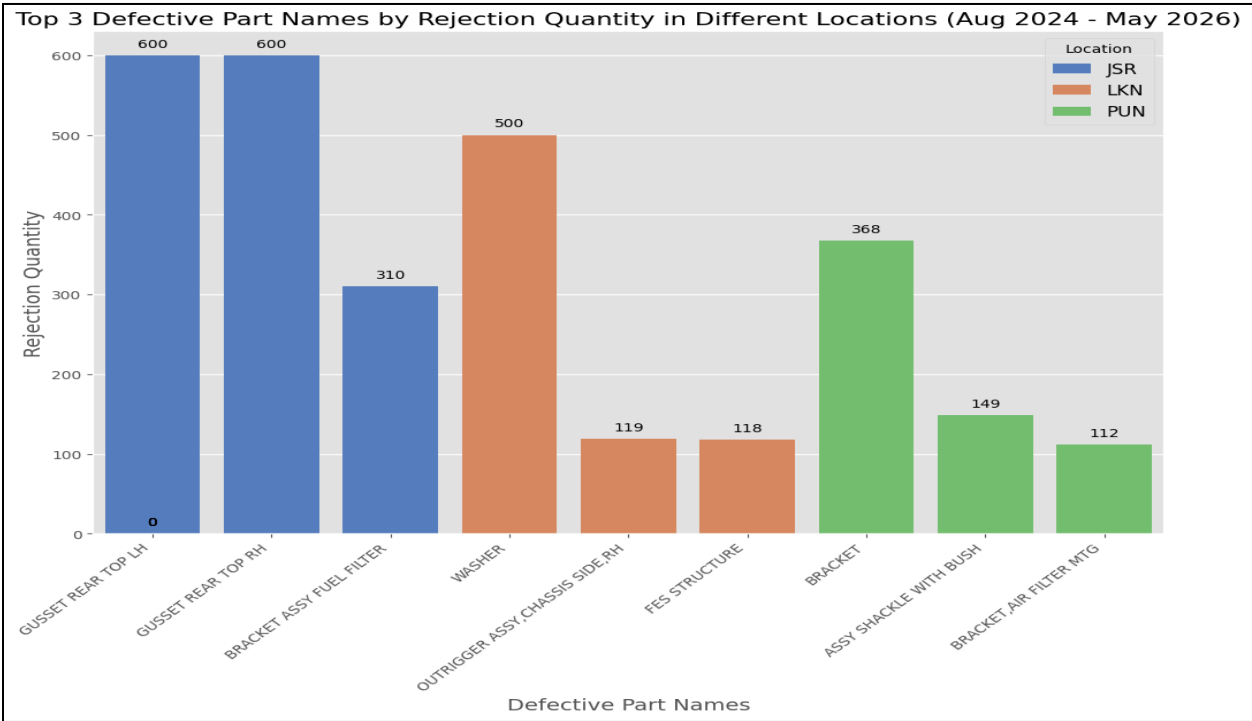


InsightEdge Observations:

- 1. **Highest Defect:** Both "Gusset Rear Top LH" and "Gusset Rear Top RH" had the highest rejection quantity at 600 each.
- 2. **Notable Defects:** The "Washer" has a significant rejection count of 500, suggesting quality concerns.
- 3. **Lower Volume:** "Bracket Assy Fuel Filter" had the least rejections at 310, indicating potential reliability.

3. Top defective parts (location-wise)

Query: Create a professional bar graph to visualize the **top 3** defective part names for each locations **['PUN', 'LKN', 'JSR']** from the month of **August 2024 to May 2026**. The x-axis should represent the defective part names for each location vertically, while the y-axis should display the rejection quantities. Each bar should correspond to a location, with distinct colors or patterns, ensure clarity and easy differentiation. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability. Label the axes and ensure all the names are clearly visible and include a descriptive title to create a professional and visually appealing presentation.

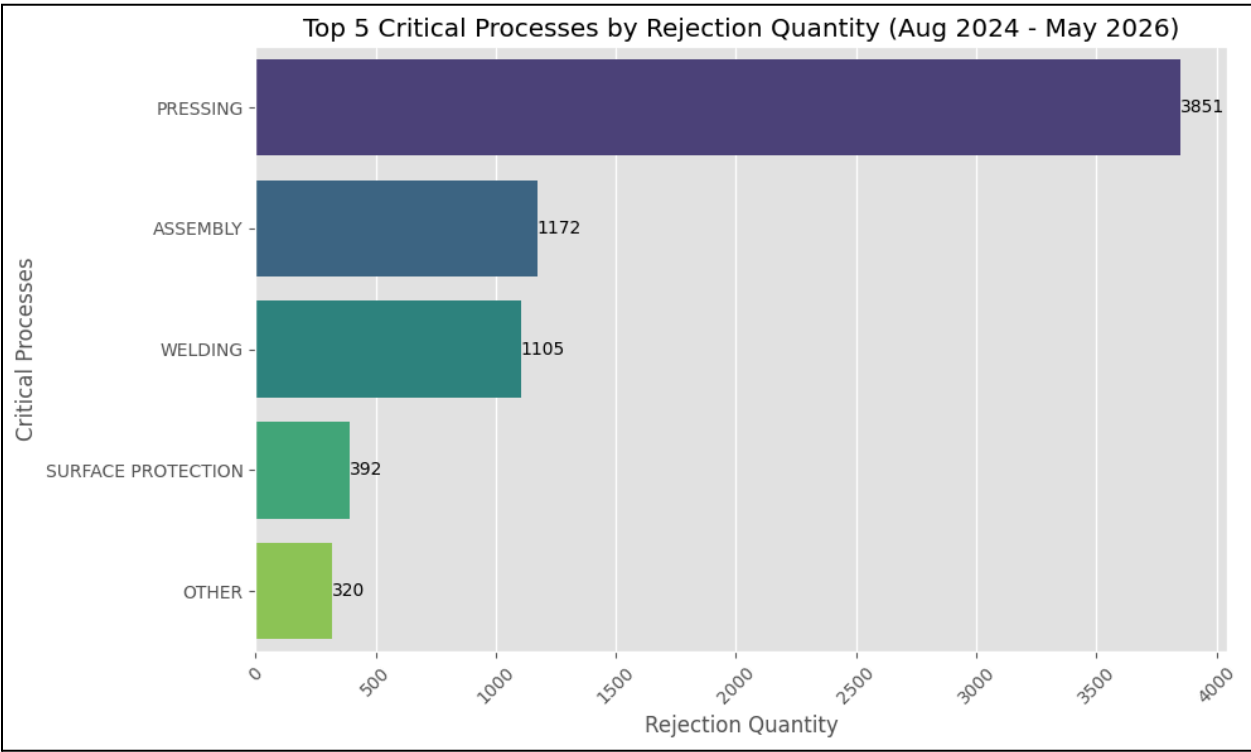


InsightEdge Observations:

- 1. **JSR Dominance:** Two parts top the rejection list at 600 units each, indicating significant quality issues.
- 2. **LKN Impact:** The part "WASHER" shows a rejection of 310 units, marking it as a critical concern.
- 3. **PUN Performance:** "ASSY SHACKLE WITH BUSH" and "BRACKET AIR FILTER MTG" have moderate rejections, suggesting potential areas for improvement.

4. Top Critical Processes

Query: Create a professional horizontal bar graph to visualize the **top 5** critical processes for all locations together **['PUN', 'LKN', 'JSR']** from the month of **August 2024 to May 2026**. The x-axis should represent the rejection quantities, while the y-axis should display the critical processes. Each bar should correspond to a specific critical process, with distinct colors or patterns, ensure clarity and easy differentiation. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability.

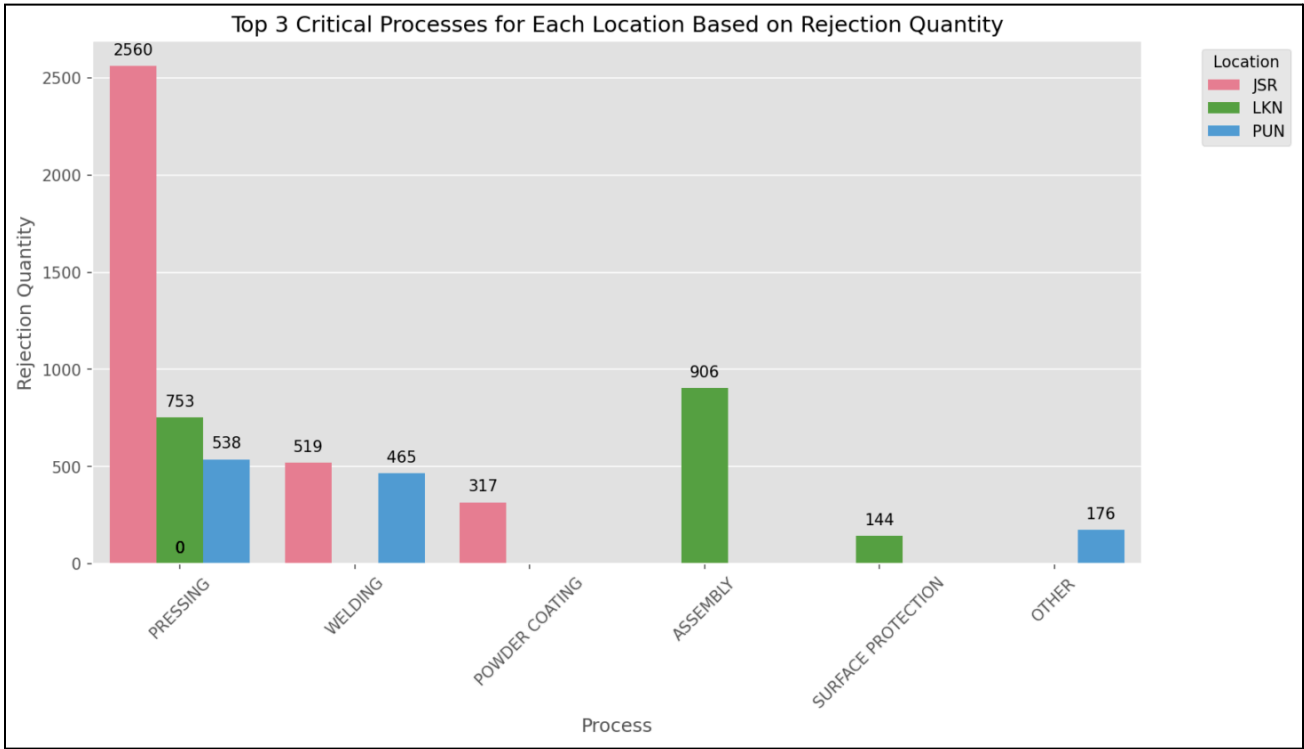


InsightEdge Observations:

- 1. **Pressing Dominance:** Pressing process leads with 3,851 rejections, significantly higher than others.
- 2. **Assembly Challenges:** Assembly follows with 1,172 rejections, indicating possible quality concerns.
- 3. **Minor Issues:** Surface protection and other processes show relatively low rejections at 392 and 320, respectively.

5. Top Critical Processes (Location-wise)

Query: Create a grouped bar chart showing the top three critical processes for each of the locations **['PUN', 'LKN', 'JSR']** from **August 2024 to May 2026**, based on total rejection quantity. Display process names on the x-axis tilted at 45 degrees and rejection quantities on the y-axis. Use distinct colors or patterns for each location and include a clear legend for easy comparison. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability

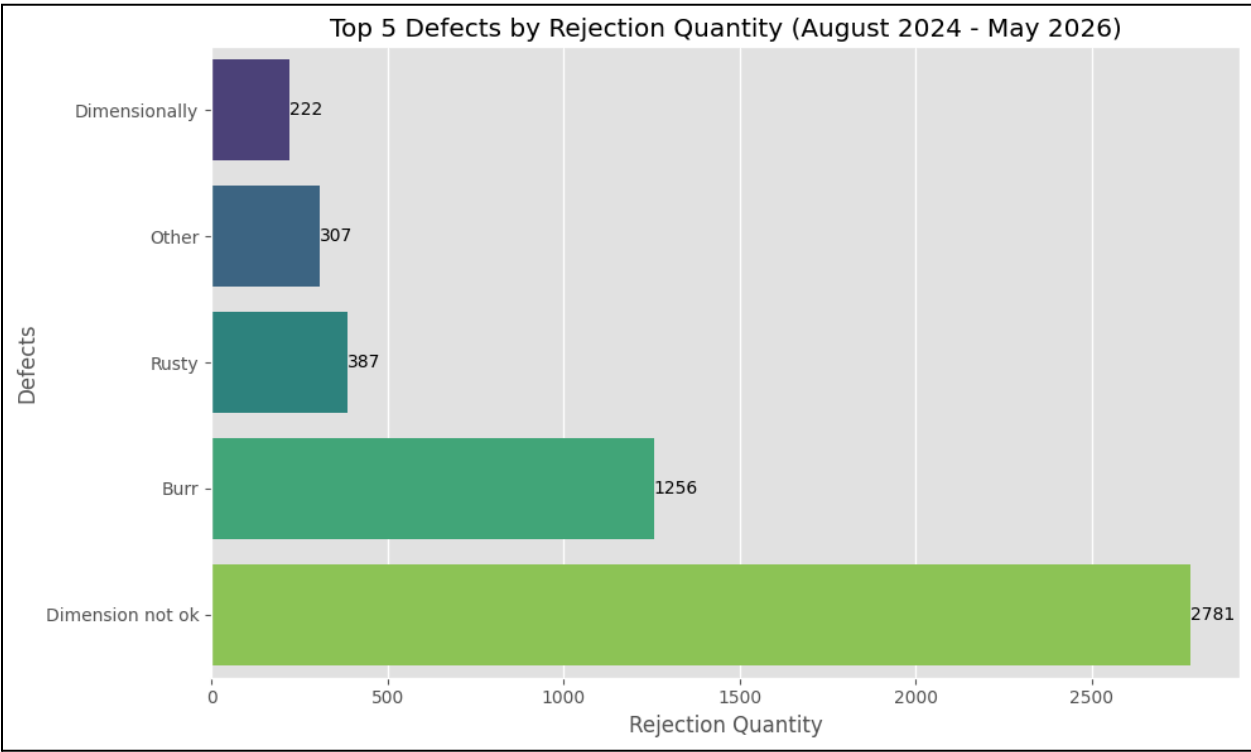


InsightEdge Observations:

- 1. **Pressing Process Dominance** :Pressing recorded the highest rejection quantity at **2560**, significantly outpacing other processes, indicating critical quality concerns in this area
- 2. **Assembly Process Rejections** :Assembly had notable rejections at **906**, suggesting it requires attention to improve quality control measures
- 3. **Minimal Rejections in Other Processes** :Processes like Welding (**519**) and Surface Protection (**144**) showed lower rejection quantities, indicating relatively better quality performance compared to Pressing and Assembly.

6. Most Impactful Root Causes

Query: Create a professional horizontal bar graph to visualize the **top 5** defects for all locations together **['PUN', 'LKN', 'JSR']** from the month of **August 2024 to May 2026**. The x-axis should represent the rejection quantities, while the y-axis should display the defects name. Each bar should correspond to a specific defect, with distinct colors or patterns, ensure clarity and easy differentiation. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability.

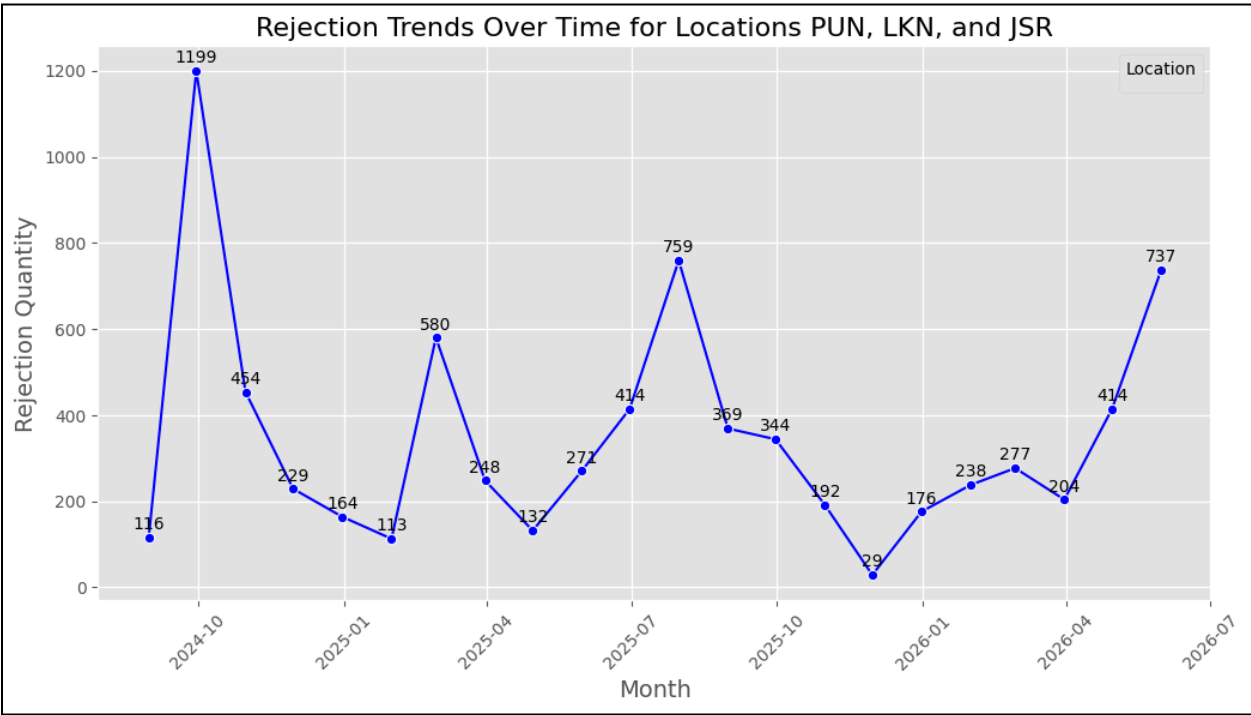


InsightEdge Observations:

- 1. **Dimension Dominance:** "Dimension not ok" represents the highest rejection quantity at 2,781.
- 2. **Significant Burrs:** "Burr" defects account for a notable 1,256 rejections, indicating a key area for improvement.
- 3. **Minor Issues:** "Dimensionality" and "Other" defects have much lower rejection counts, at 222 and 307 respectively.

7. Time-Based Rejection Trends

Query: Create a line chart for the rejection trends over time for all locations together ['PUN', 'LKN', 'JSR'], starting from August 2024 to May 2026, by resampling the data on a monthly basis. The X-axis should represent the months while the Y-axis should represent the rejection quantities. Include annotations to highlight significant peaks or drops in rejection trends during this period. Label the axes and ensure all the names are clearly visible and include a descriptive title to create a professional and visually appealing presentation. Annotate each points on line chart.

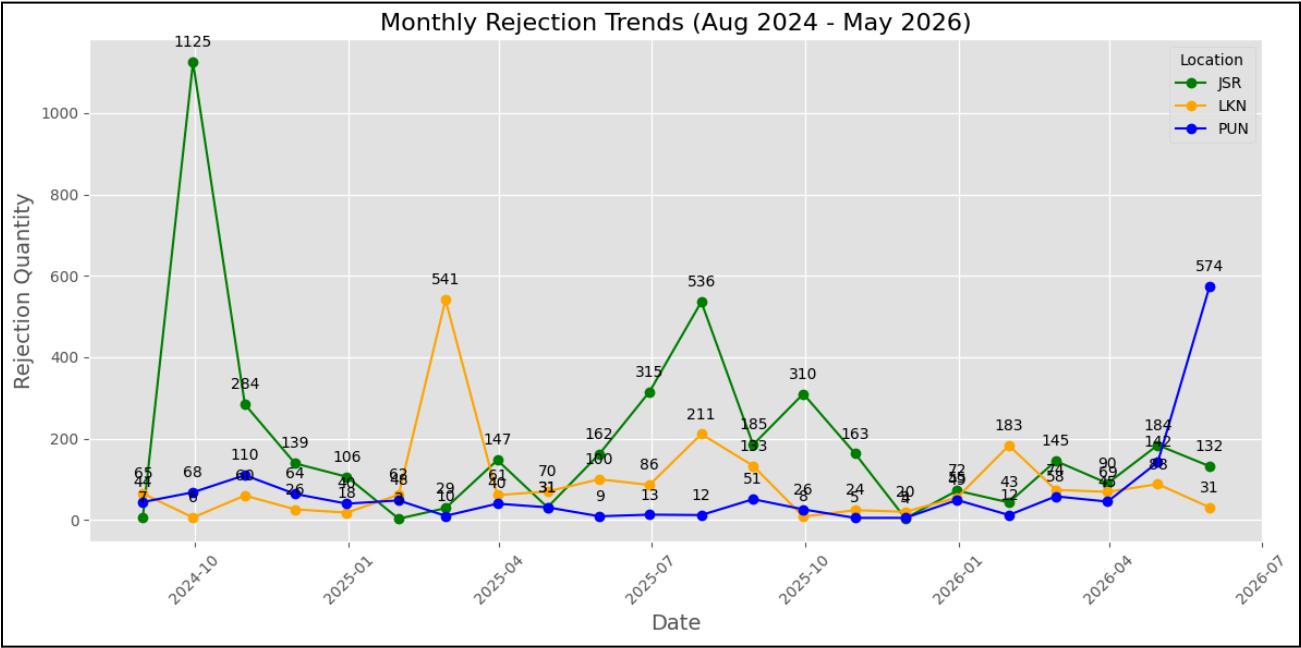


InsightEdge Observations:

- 1. **Initial Peak:** Rejections reached a high of 1199 in October 2024 before showing significant fluctuation.
- 2. **Midpoint Drop:** By January 2025, rejections fell to a low of 113, indicating a substantial decline in the middle trend.
- 3. **Recent Recovery:** Rejections increased again to 737 by July 2026, suggesting a potential upward trend.

8. Time-Based Rejection Trends (Location-wise)

Query: For each in ['PUN', 'LKN', 'JSR'], visualize the rejection trends over time starting from **August 2024 to May 2026** by resampling the data on a monthly basis. Use a multi-line chart with distinct colors for each location in ['PUN', 'LKN', 'JSR'] and a clear legend for each location. Add annotations of rejection quantities for significant peaks and drops to highlight inflection points, ensuring the chart is visually clean and readable. Annotate each points on line chart.

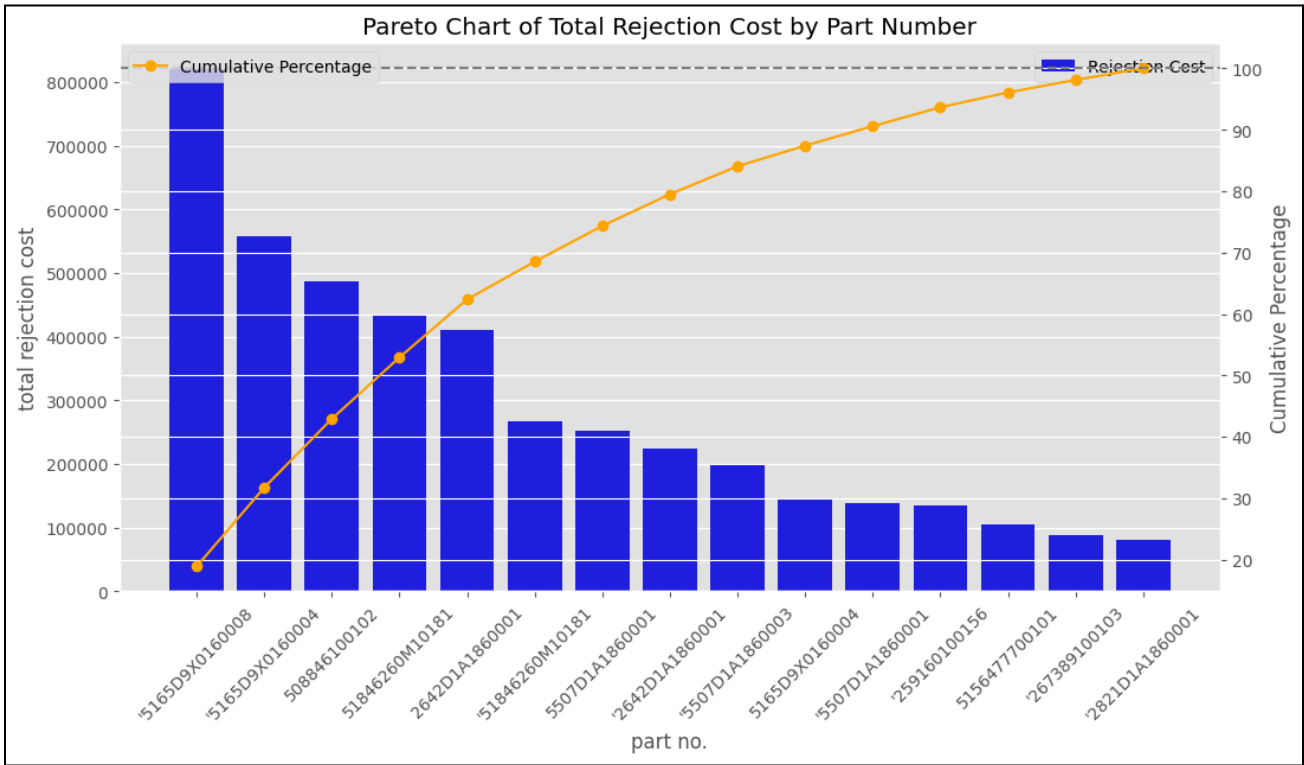


InsightEdge Observations:

- 1. **JSR Peak:** JSR rejections peaked at 1125 in October 2024 before experiencing a significant drop.
- 2. **LKN Fluctuations:** LKN had a notable rejection peak at 541 in March 2025, followed by a decrease to 90.
- 3. **PUN Consistency:** PUN showed consistent low rejection values, rarely exceeding 300 throughout the observed period.

9. Pareto analysis for Total Rejection Cost

Query: Create a Pareto chart for the period **August 2024 to May 2026** across **['PUN', 'LKN', 'JSR']**, showing the **top 15** part numbers ranked by total rejection cost. Display bars for rejection cost and a cumulative percentage line on a secondary axis. Rotate x-axis labels exactly 45 degrees. Do not annotate bars or line points with any values. Strictly rotate the x-axis labels by 45 degrees

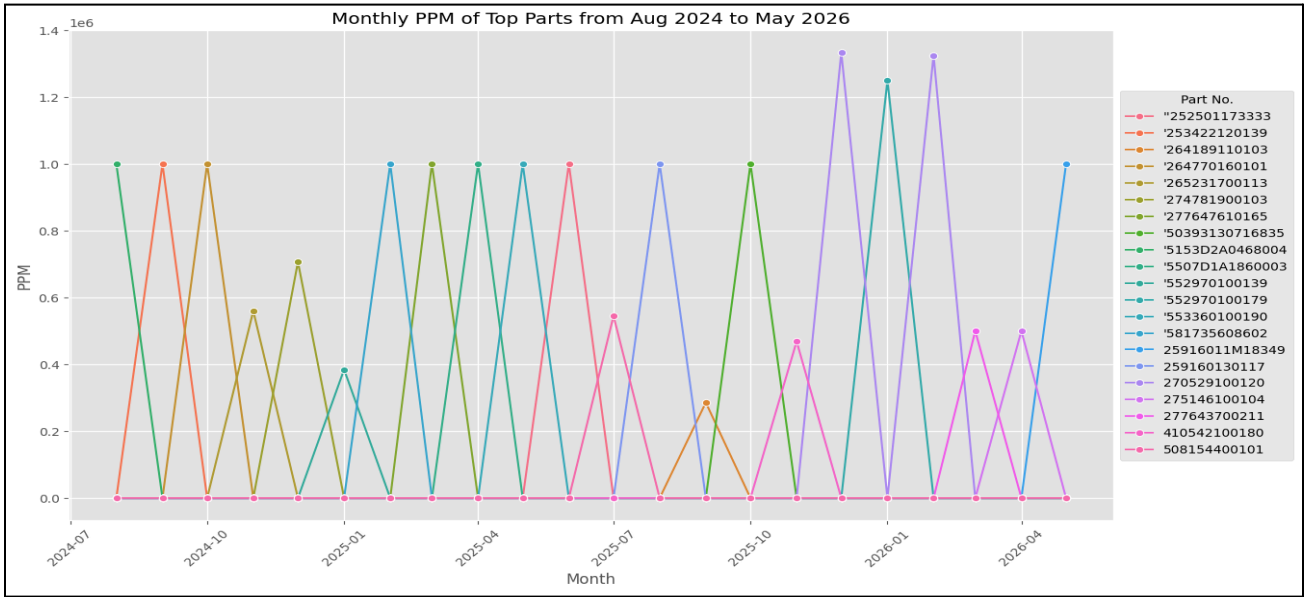


InsightEdge Observations:

- 1. **Highest Cost:** One part accounts for over 80,000 in total rejection costs, dominating the total.
- 2. **Gradual Increase:** The cumulative percentage line suggests that a few parts contribute significantly to overall rejection costs.
- 3. **Long Tail:** Many parts have relatively low rejection costs, indicating potential areas for improvement.

10. PPM Trends

Query: From **August 2024 to May 2026**, identify the part number with the highest PPM each month, where ****PPM = (Rejection Qty × 1,000,000) / Sale Qty****. Then, for each selected part number, compute its PPM across all months, using 0 where data is missing. Organize the results in a table with part numbers as rows and months as columns. Finally, plot a multi-line chart with months (as datetime) on the X-axis, PPM on the Y-axis, and each line representing a part number.Keep the legend on right side showing the complete chart



InsightEdge Observations:

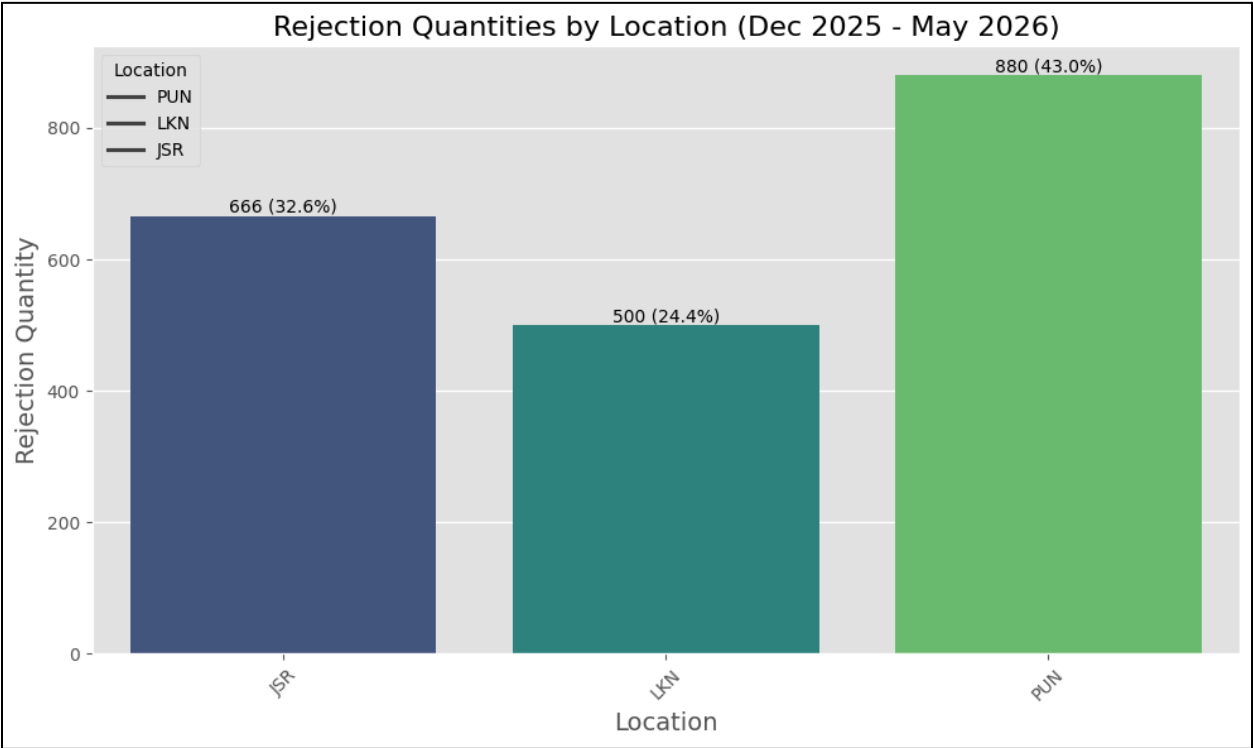
- High PPM Fluctuations:** The graph shows sharp fluctuations in PPM values across different months, indicating inconsistent quality performance over time.
- Major PPM Spikes:** Significant spikes are observed during **June 2025, August 2025, and October 2025**, where PPM values reached extremely high levels, highlighting critical quality concerns.
- Part Number Impact:** Certain part numbers contributed disproportionately to monthly PPM peaks, suggesting the need for targeted root cause analysis for specific components.
- Stable Low-PPM Periods:** Several months maintained relatively low PPM values, reflecting periods of stable manufacturing quality and improved defect control

Agenda 2:

Trend analysis for Last 6 months
(excluding Current Month)

1. Location-Wise Summary

Query: Create a bar graph to visualize the rejection quantities for location ['PUN', 'LKN', 'JSR'] for the months from December 2025 to May 2026. The x-axis should represent the locations ['PUN', 'LKN', 'JSR'], while the y-axis should display the rejection quantities. Each bar should represent the total rejection quantity for a specific location, color-coded to ensure clarity. Add a legend to identify the locations and annotate each bar with the exact rejection quantity and its percentage contribution to the total sum of rejection quantities, assuming the sum equals 100%. Ensure the annotations are clear and placed near the bars for readability. Label the axes and include a title for a professional and clean presentation.

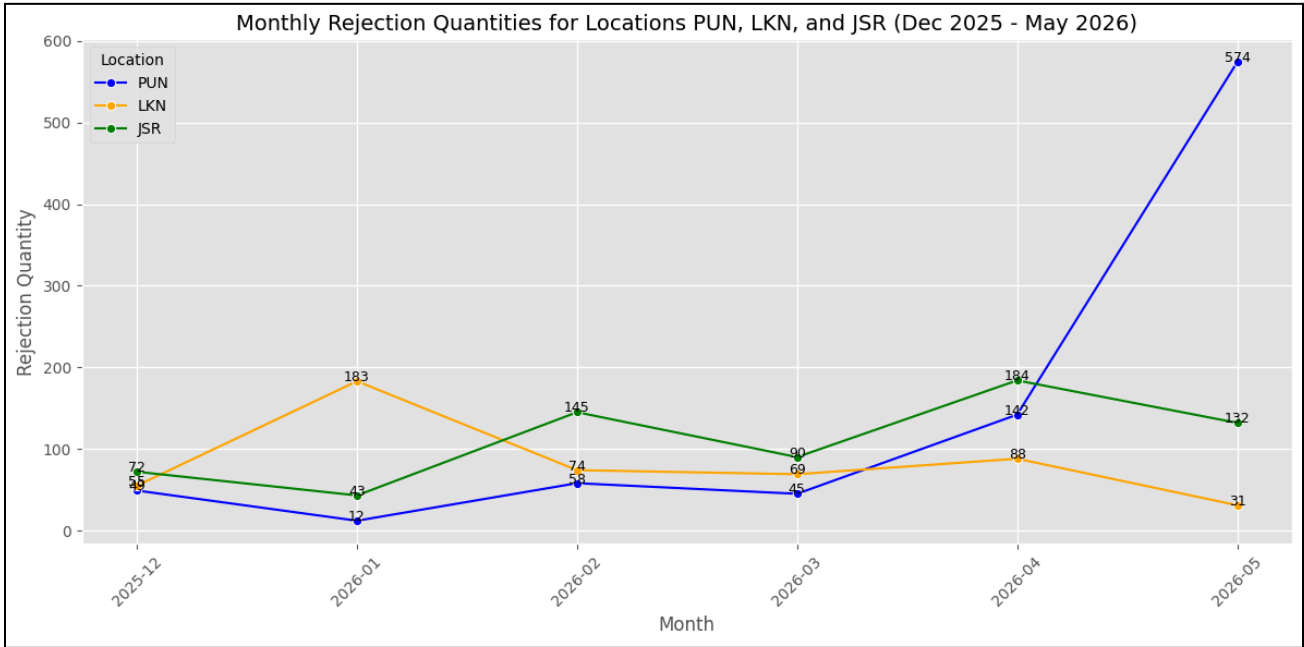


InsightEdge Observations:

- 1. **Highest Rejection:** PUN had the highest rejection quantity at 880, accounting for 43.0% of total rejections.
- 2. **Moderate Impact:** JSR's rejections totaled 666, representing 32.6% of total rejections.
- 3. **Lowest Rejections:** LKN recorded the lowest quantity at 500, which is 24.4% of the overall totals.

2. Trend Analysis of Rejection Quantities

Query: Create a professional multi-line chart to visualize the rejection quantities for locations **['PUN', 'LKN', 'JSR']** across the months from **December 2025 to May 2026**. The x-axis should represent the months in strict chronological order based on the selected date range. The y-axis should display the rejection quantities for each location. Each location should be represented by a distinct line with appropriate markers and colors to ensure clarity and distinction. Include a legend to identify the locations **['PUN', 'LKN', 'JSR']**. The chart should have labeled axes and a title to maintain a clean and professional appearance. Avoid emphasizing specific points or adding additional highlights.

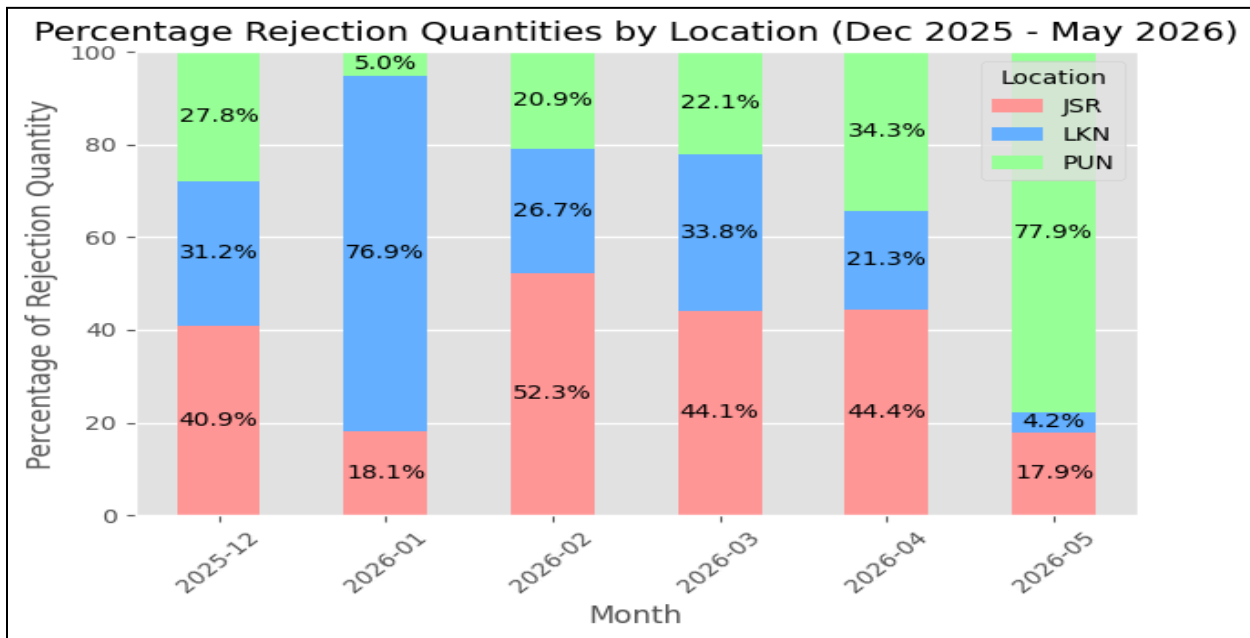


InsightEdge Observations:

- 1. **PUN Peak:** Rejection quantities for PUN peaked notably in May 2026, reaching around 300 units.
- 2. **LKN Sharp Increase:** LKN recorded a significant spike of 183 in January 2026, highlighting a major quality or process issue during that period.
- 3. **JSR Highest Variability:** JSR showed major fluctuations in rejection quantities, peaking at 184 in April 2026 and again at 145 in February 2026, indicating inconsistent quality performance..
- 4. **Overall Trend Variation:** Rejection quantities varied across all three locations during the period, suggesting changing operational conditions and the need for continuous quality monitoring.

3. Monthly Contributions to Total Rejections

Query: Create a segmented bar graph to visualize the percentage rejection quantities for each month from **December 2025 to May 2026**, divided by location **['PUN', 'LKN', 'JSR']**. The y-axis should represent percentages ranging from 0% to 100%, while the x-axis should display the months in strict chronological order between **December 2025** and **May 2026**. Each bar should represent the total rejection quantity for a specific month, segmented into portions corresponding to the percentage contributions of each location, ensuring the sum of all segments for a given month equals 100%. Distinct colors or patterns should be used to represent each location, and a legend should be included to identify them.

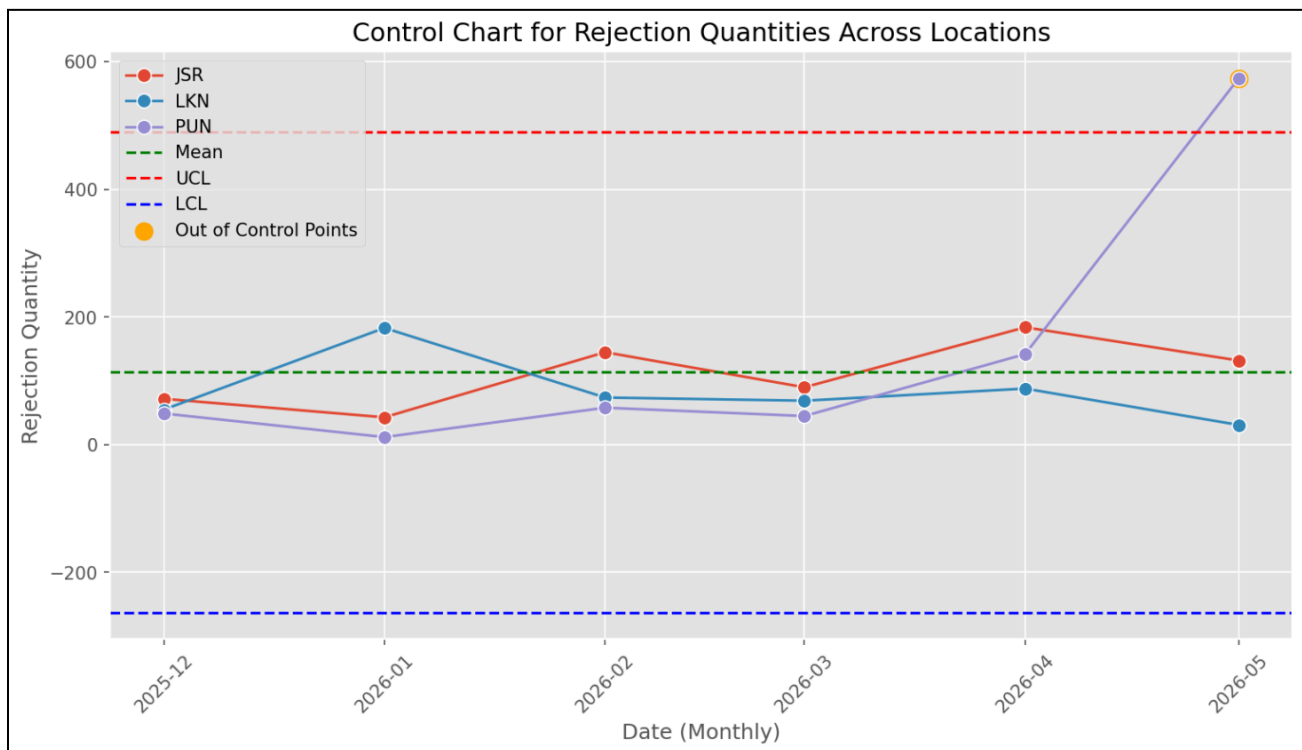


InsightEdge Observations:

- JSR Increase:** JSR rejections peaked at 76.9% in January before declining to 21.3% by May.
- LKN Stability:** LKN maintained a steady rejection percentage around 44%, suggesting consistent performance.
- PUN Drop:** PUN saw a significant drop from 34.3% in January to only 4.2% in May, indicating improved conditions.

4. Control Chart for Total Rejections

Query: Create a control chart for rejection quantities across locations ['PUN', 'LKN', 'JSR'] from **December 2025 to May 2026**. Plot months in strict chronological order on the x-axis while the y-axis should display the rejection quantities. Calculate the mean rejection quantity and determine the upper control limit (UCL) and lower control limit (LCL) as three standard deviations above and below the mean, respectively. Display these control limits using dashed lines: green for the mean, red for the UCL and blue for the LCL. Use distinct markers for individual data points for each date on the chart to ensure clarity. Highlight any data points that fall outside the control limits in a different color for easy identification.

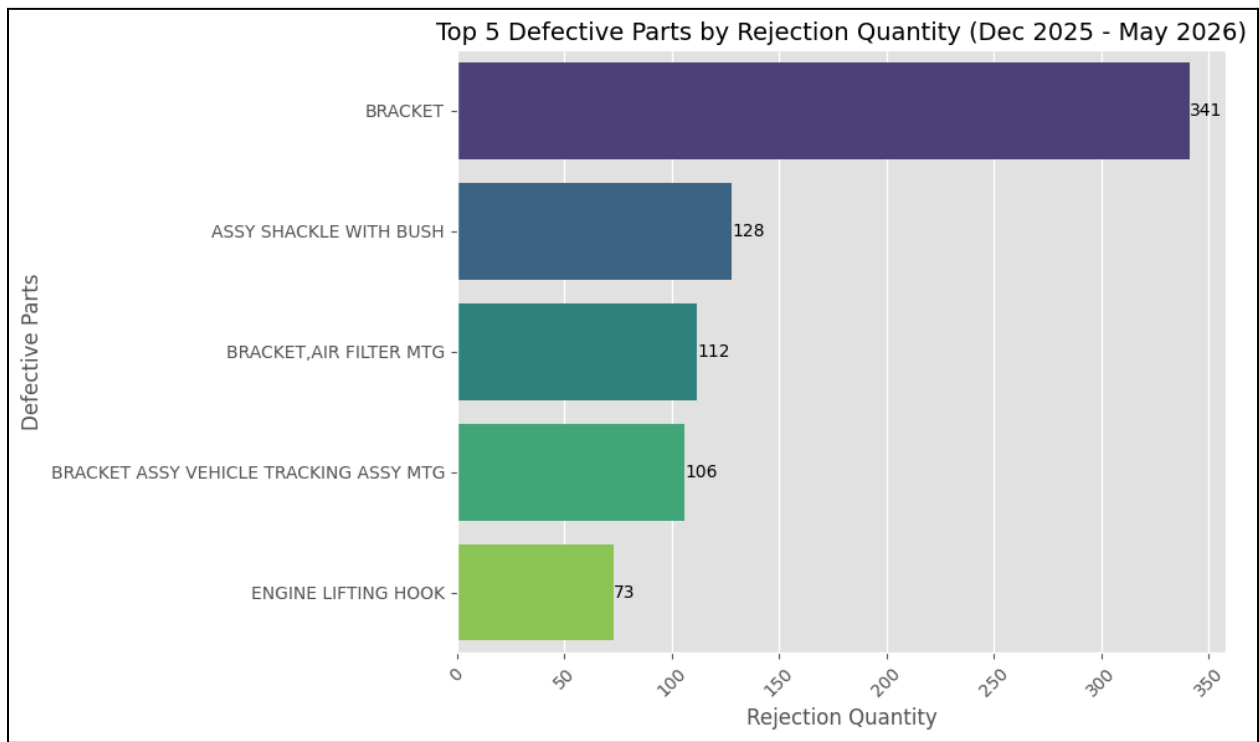


InsightEdge Observations:

- JSR Variability:** JSR shows significant fluctuations, with peaks reaching nearly 300 rejections on certain dates.
- LKN Stability:** LKN displays consistent low rejection rates, rarely exceeding 50 rejections throughout the period.
- PUN Surge:** PUN has a notable spike, reaching over 600 rejections at isolated instances, indicating irregular patterns.

5. Top Defective Parts

Query: Create a professional horizontal bar graph to visualize the **top 5** defective parts for all locations **['PUN', 'LKN', 'JSR']** for the months from **December 2025 to May 2026**. The x-axis should represent the rejection quantities, while the y-axis should display the names of the defective parts. Each bar should correspond to a specific defective part, with distinct colors or patterns ensuring clarity and easy differentiation. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability. Label the axes and ensure all names are clearly visible, and include a descriptive title to create a professional and visually appealing presentation.

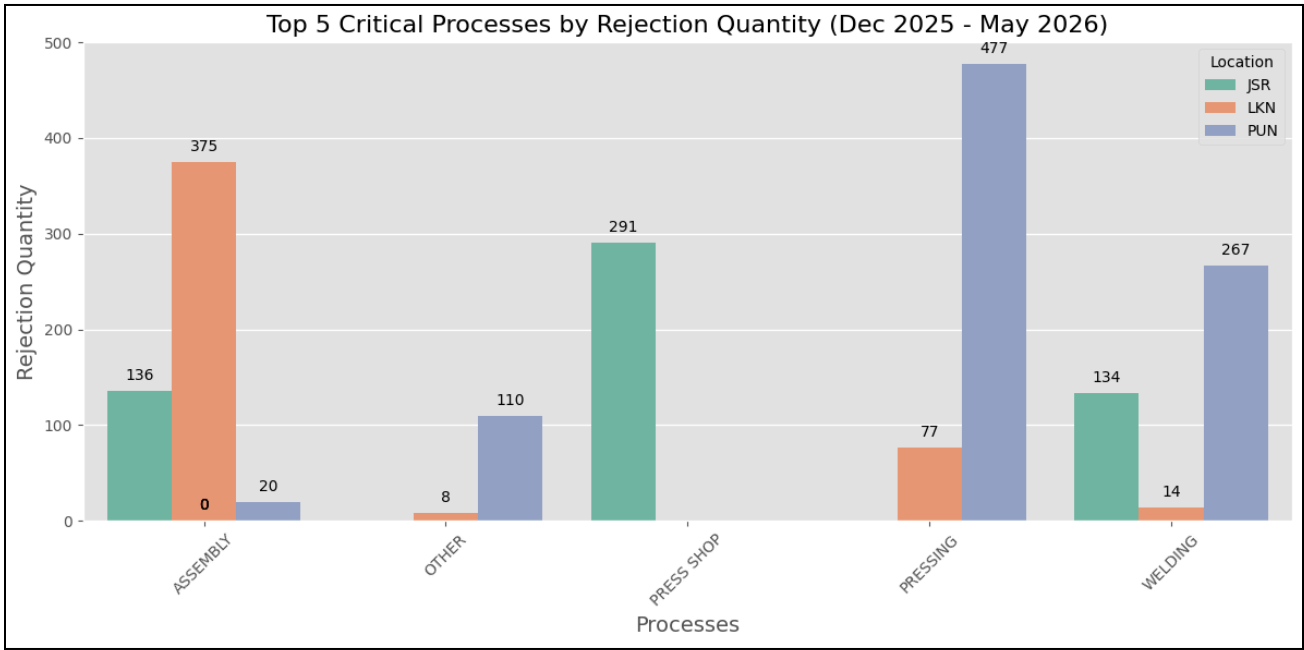


InsightEdge Observations:

- 1. **Bracket Dominance:** The "Bracket" part accounts for 341 rejections, significantly surpassing others.
- 2. **Secondary Issues:** "Assy Shackle with Bush" and "Bracket, Air Filter MTG" follow, with 128 and 112 rejections, respectively.
- 3. **Minor Defects:** "Engine Lifting Hook" shows the least rejections at 73, indicating a lower defect rate.

6. Top Critical Processes

Query: Create a professional bar graph to visualize the **top 5** critical processes for all locations **['PUN', 'LKN', 'JSR']** during the months from **December 2025 to May 2026**. The x-axis should represent the names of the processes, while the y-axis should display the rejection quantities. Individual bars should be displayed by location, with distinct colors or patterns used to represent each location, ensuring clarity and easy differentiation. Include a legend to identify the locations, and ensure that all names and segments are clearly visible. Annotate each bar with the exact rejection quantity, positioning the annotations appropriately for readability. Label the axes and include a descriptive title to maintain a professional and visually appealing presentation.

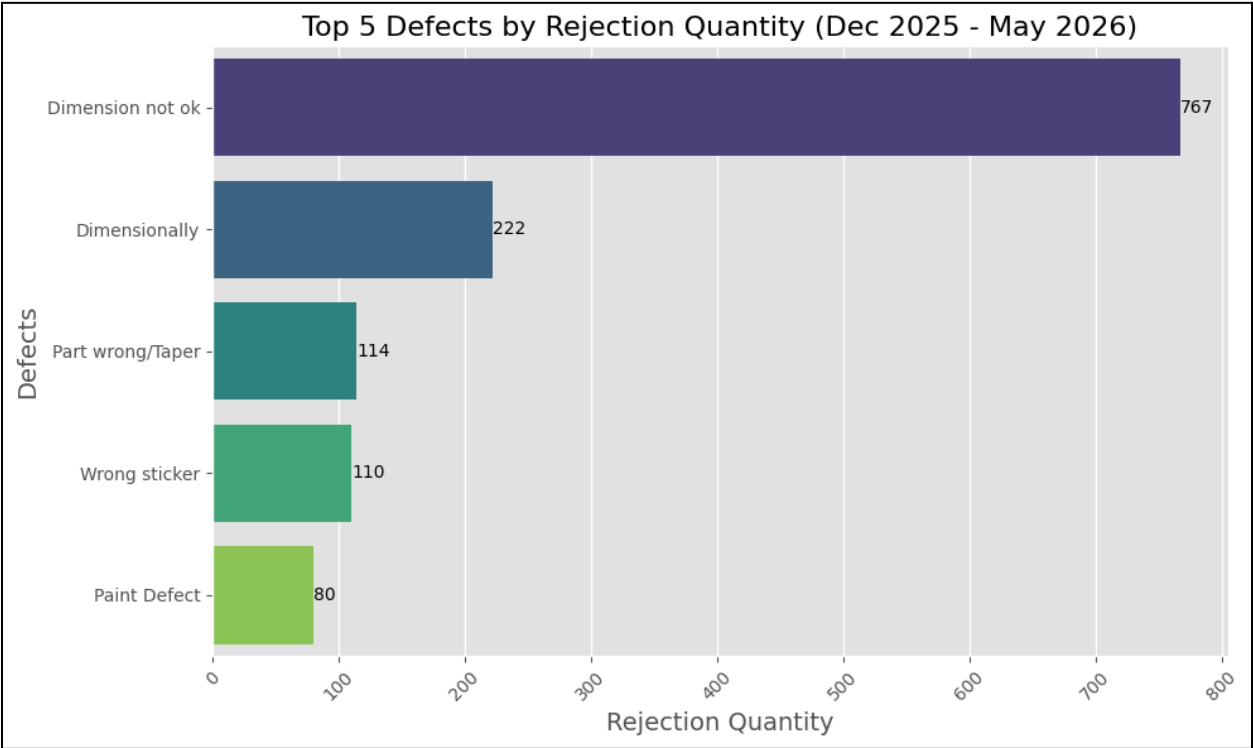


InsightEdge Observations:

- LKN's Dominance :** LKN has the highest rejection quantity with **375** for **ASSEMBLY** process, significantly leading other locations.
- JSR's Consistency :** JSR demonstrated moderate rejection levels with consistent quantities, peaking at **291** for **Press Shop** process, suggesting stable but non-optimal performance
- PUN's Lower Impact :** PUN had the least rejections at **20** for **Assembly** and **110** for **OTHER** process, implying better quality control, though further improvements could enhance overall performance.

7. Most Impactful Root Causes

Query: Create a professional horizontal bar graph to visualize the **top 5** defects for all locations **['PUN', 'LKN', 'JSR']** for the months from **December 2025 to May 2026**. The x-axis should represent the rejection quantities, while the y-axis should display the names of the defects. Each bar should correspond to a specific defect, with distinct colors or patterns ensuring clarity and easy differentiation. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability. Label the axes and ensure all names are clearly visible, and include a descriptive title to create a professional and visually appealing presentation.

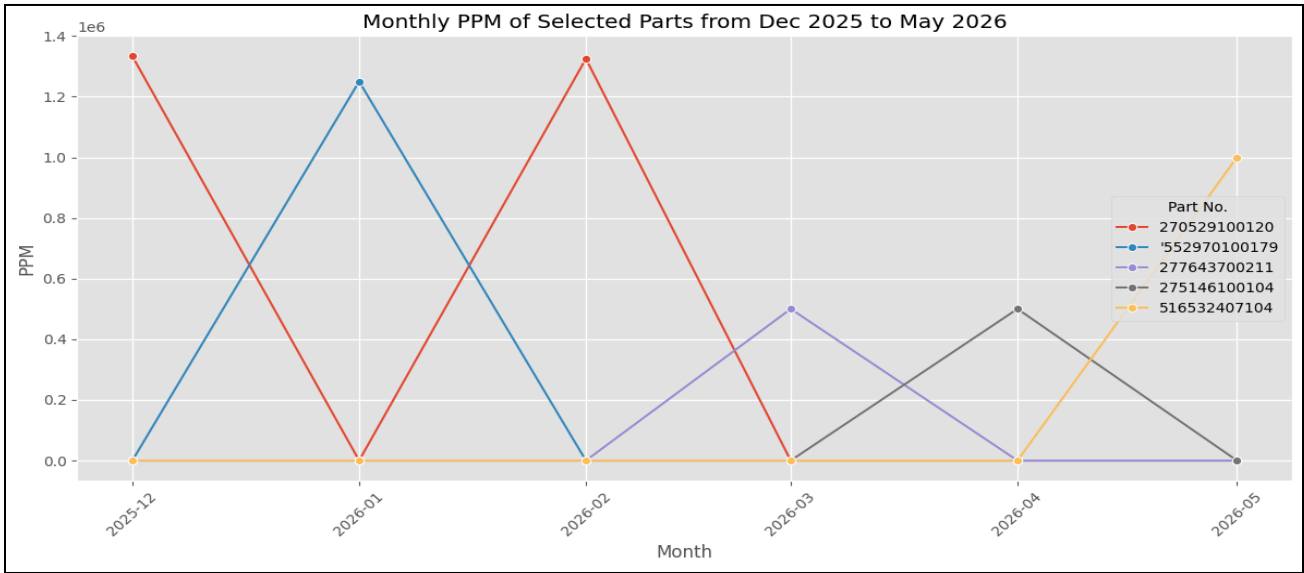


InsightEdge Observations:

- 1. **Major Defect:** "Dimension not ok" accounts for 767 rejections, dominating the chart.
- 2. **Secondary Issue:** "Dimensionally" has notable rejections at 222, indicating a significant concern.
- 3. **Minor Defects:** Other categories like "Part wrong/Taper," "Wrong sticker," and "Paint Defect" have lower rejection counts, suggesting less critical issues.

8. PPM Trends

Query: Follow the instructions step by step: For every month from **December 2025 to May 2026**, select the part number with the highest PPM, selecting a single part number for each month. PPM = (Rejection Quantity × 1,000,000) / Sale Quantity. For every selected part number, calculate its PPM across all months in the selected period. If no data is found for a month, assign a value of 0. Structure the results as a table where each row represents a part number and each column represents a month, with values representing PPM. Generate a multi-line chart using the data. The x-axis represents months (convert to datetime before plotting), and the y-axis represents PPM. Each line represents a part number.

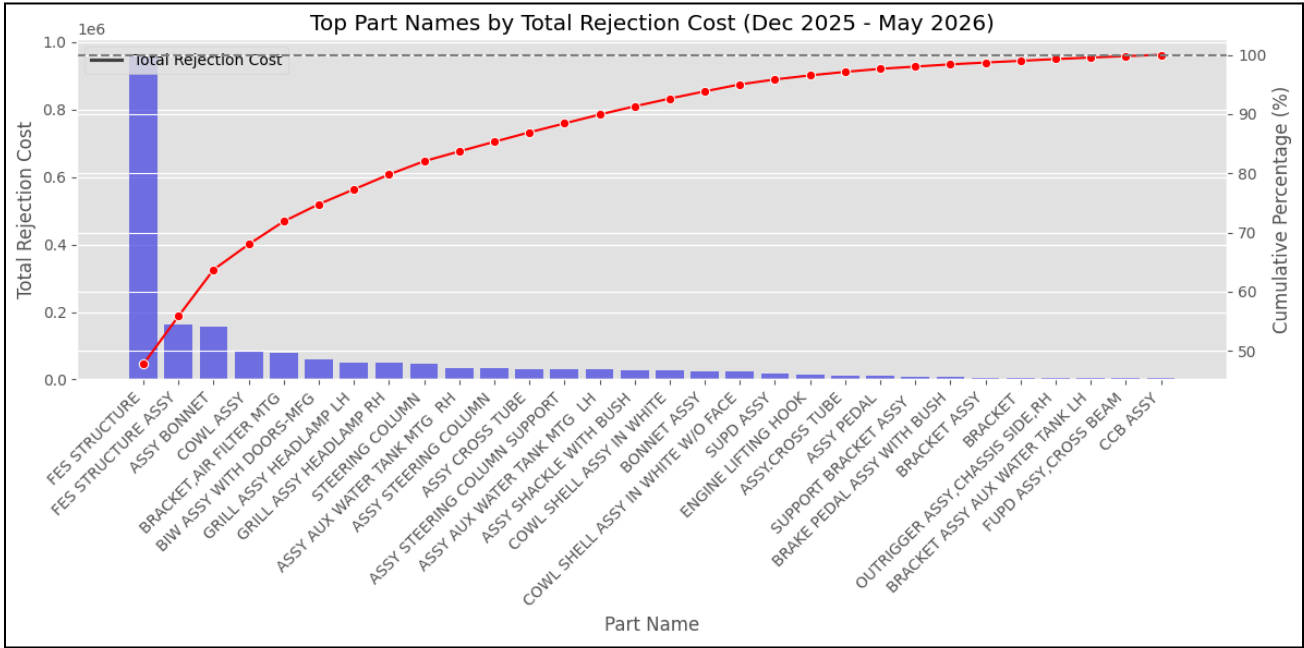


InsightEdge Observations:

- 1. **Major PPM Spikes:** Certain part numbers showed extremely high PPM spikes in December 2025 and February 2026, indicating critical quality issues that require immediate investigation.
- 2. **Fluctuating Quality Performance:** PPM values varied significantly across months, reflecting inconsistent production quality for some part numbers.
- 3. **Stable Low-PPM Parts:** Several part numbers maintained consistently low PPM levels throughout the period, suggesting stable manufacturing performance.
- 4. **Recurring High PPM Trends:** Repeated spikes for specific part numbers indicate persistent process or material-related problems that need corrective action.

9. Pareto analysis for Total Rejection Cost

Query: Pareto chart showing top part names by total rejection cost with cumulative percentage line for the period from **December 2025 to May 2026** across locations **['PUN', 'LKN', 'JSR']**. Include labels only for the **top 30** parts. Do not annotate points with rejection quantity or percentage. Strictly Rotate x-axis labels exactly at 45 degrees with names assigned to corresponding bars.



InsightEdge Observations:

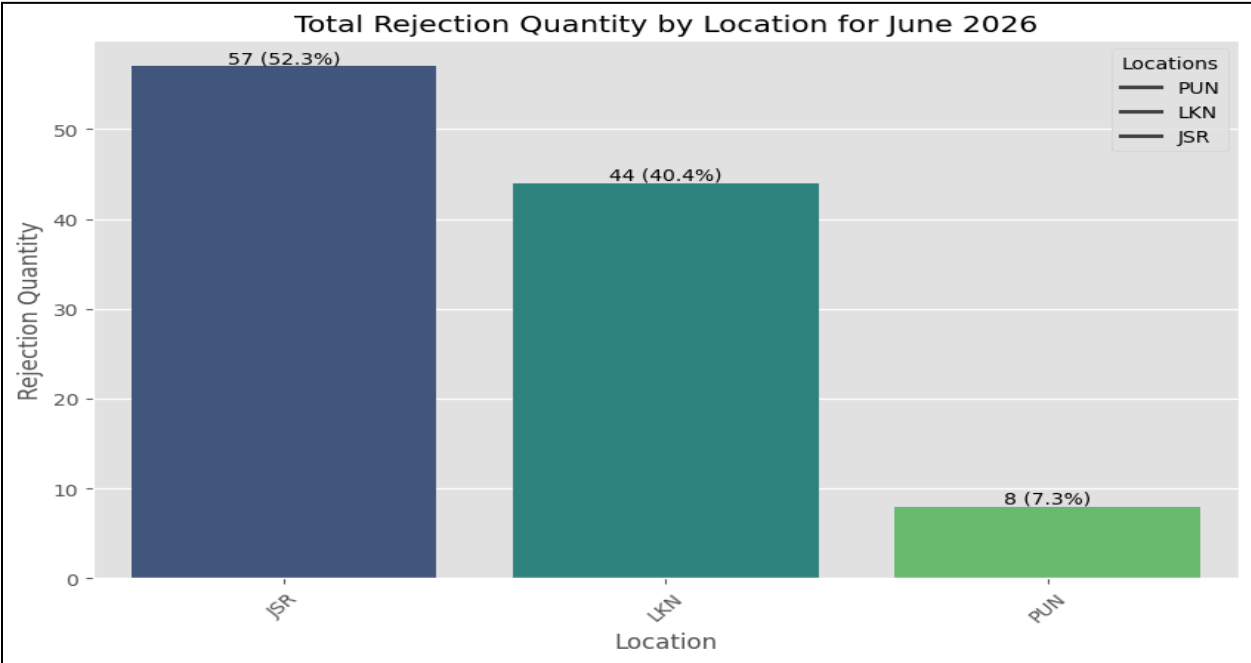
- 1. **Dominant Cost:** One part name accounts for a significant majority of total rejection costs, emphasizing a critical focus area.
- 2. **Cumulative Growth:** The cumulative percentage of costs rises steeply, indicating that few parts drive most rejections.
- 3. **Extensive Spread:** A wide range of part names has relatively low rejection costs, suggesting potential for cost reduction initiatives.

Agenda 3:

Summary of Current Month

1. Location-Wise Summary

Query: Create a bar graph to visualize the rejection quantities for three locations ['PUN', 'LKN', 'JSR'] for the month of June 2026. The x-axis should represent the locations ['PUN', 'LKN', 'JSR'], while the y-axis should display the rejection quantities. Each bar should represent the total rejection quantity for a specific location, color-coded to ensure clarity. Add a legend to identify the locations and annotate each bar with the exact rejection quantity and its percentage contribution to the total sum of rejection quantities, assuming the sum equals 100%. Ensure the annotations are clear and placed near the bars for readability. Label the axes and include a title for a professional and clean presentation.

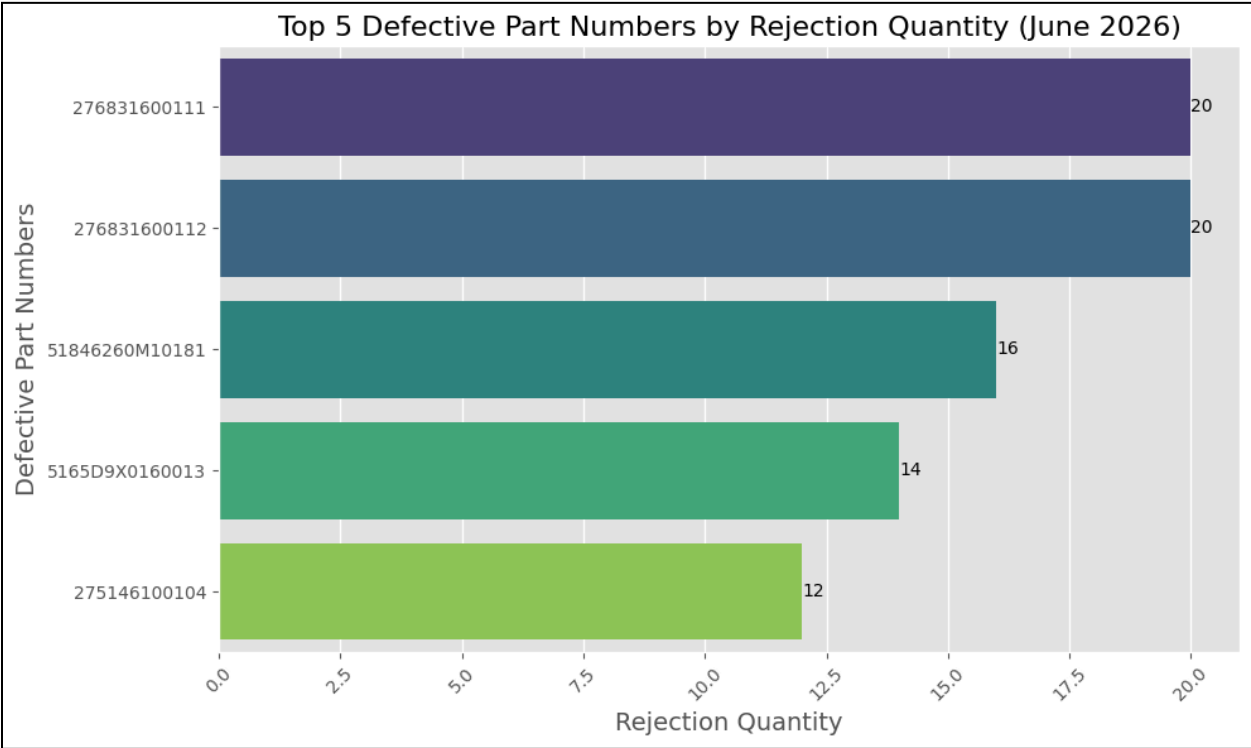


InsightEdge Observations:

- 1. **JSR Dominance:** JSR has the highest rejections at 57, representing 52.3% of total rejections.
- 2. **LKN Significance:** LKN follows with 44 rejections, accounting for 40.4% of total rejections.
- 3. **PUN Minimal:** PUN shows the lowest rejections at 8, contributing only 7.3% to the total.

2. Top Defective Parts (Overall)

Query: Create a professional bar graph to visualize the **top 5** defective part numbers for all locations **['PUN', 'LKN', 'JSR']** for the month of **June 2026**. The x-axis should represent the rejection quantities, while the y-axis should display the defective part numbers. Each bar should correspond to a specific defective part number, with distinct colors or patterns ensuring clarity and easy differentiation. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability. Label the axes and ensure all the numbers are clearly visible, and include a descriptive title to create a professional and visually appealing presentation.

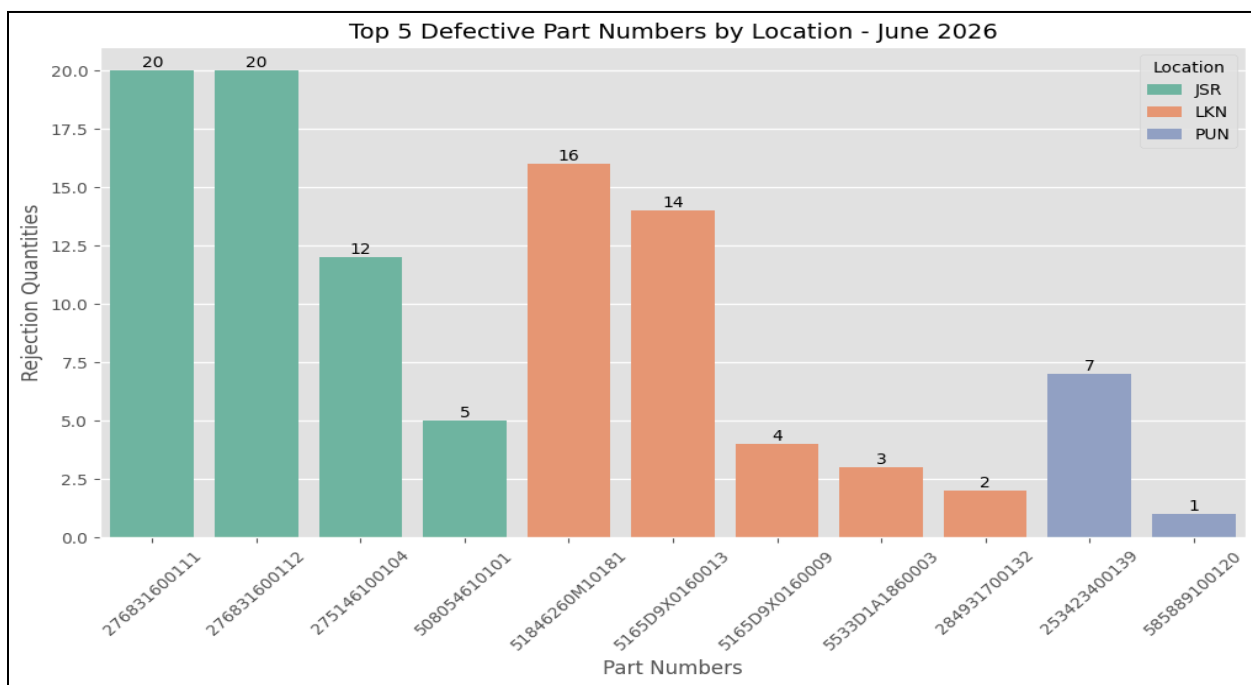


InsightEdge Observations:

- 1. **Top Defectives:** Two part numbers share the highest rejection quantity at 20 each.
- 2. **Notable Drop:** The next parts have significantly lower rejections, with the highest at 16.
- 3. **Consistent Issues:** The bottom parts show persistent defects, with rejections ranging from 12 to 14.

3. Top Defective Parts (Location-wise)

Query: Create a professional vertical column bar graph to visualize the **top 5** defective part numbers for each of the locations **['PUN', 'LKN', 'JSR']** for the month of **June 2026**. The y-axis should represent the rejection quantities, while the x-axis should display the defective part numbers. Each bar should correspond to a specific defective part number, with distinct colors or patterns ensuring clarity and easy differentiation. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability. Label the axes and ensure all the numbers are clearly visible, and include a descriptive title to create a professional and visually appealing presentation.

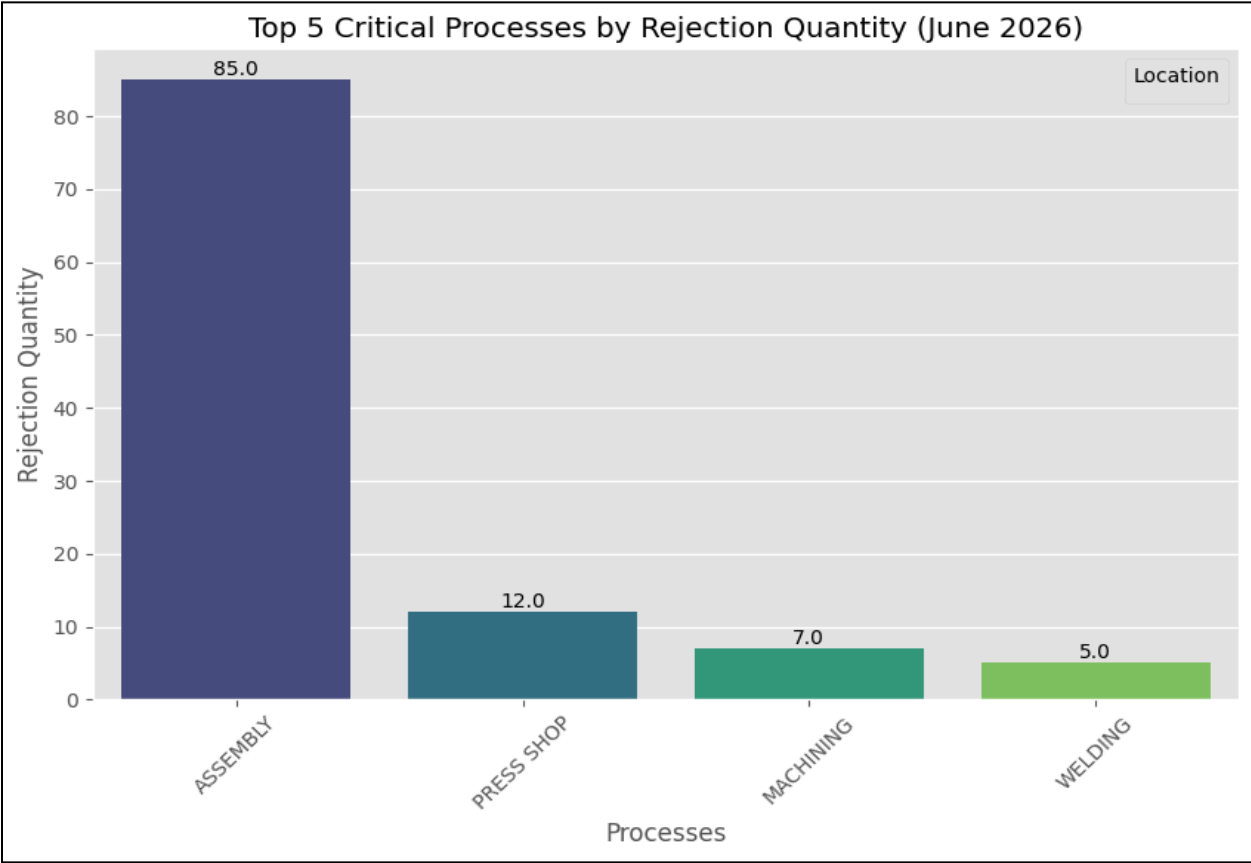


InsightEdge Observations:

- JSR Dominance:** The highest rejection quantity is 20 for two part numbers, indicating significant issues at this location.
- LKN Peak:** LKN shows a notable peak at 16 rejections for a specific part number, suggesting targeted quality concerns.
- PUN Low:** PUN has the lowest rejection quantity of 1, reflecting minimal defect issues compared to other locations.

4. Top Critical Processes (Overall)

Query: Create a professional bar graph to visualize the **top 5** critical processes across all locations **['PUN', 'LKN', 'JSR']** during the month of **June 2026**. The x-axis should represent the names of the processes, while the y-axis should display the rejection quantities. Annotate each bar with rejection quantities.

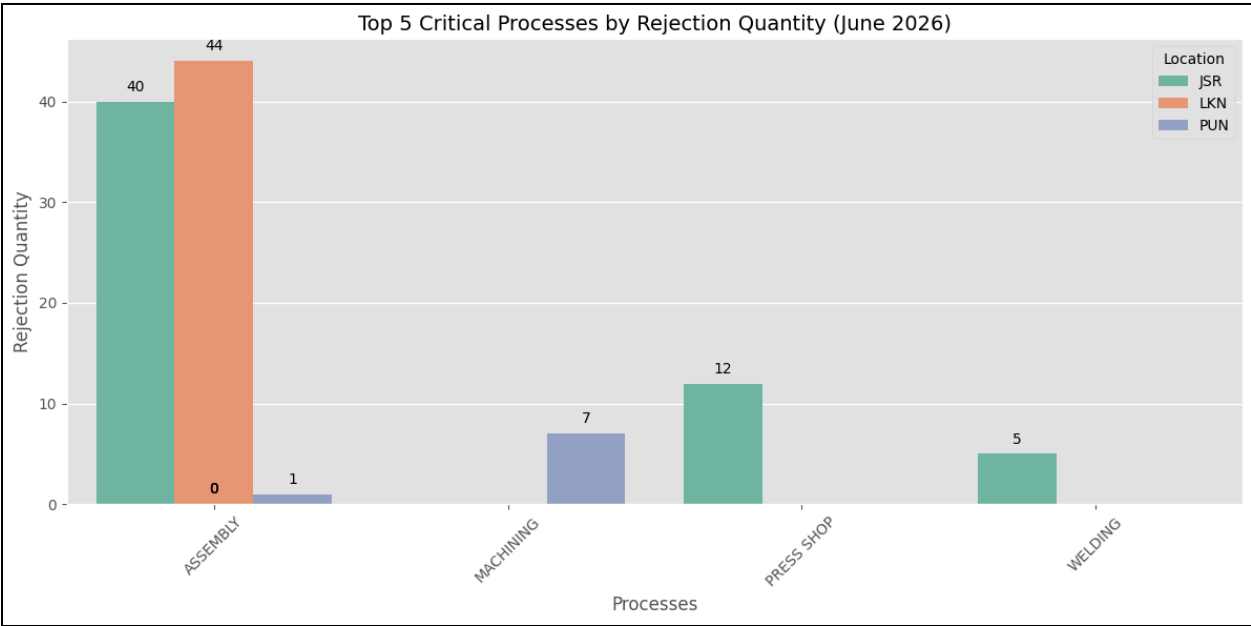


InsightEdge Observations:

- 1. **Assembly Dominance:** The assembly process has the highest rejection quantity at 85, significantly overshadowing others.
- 2. **Minor Rejections:** Press shop, machining, and welding show low rejection quantities of 12, 7, and 5, respectively.
- 3. **Focus Area:** Assembly needs immediate attention to reduce rejection rates and improve overall quality.

5. Top Critical Processes (Location-wise)

Query: Create a professional bar graph to visualize the **top 5** critical processes for all locations **['PUN', 'LKN', 'JSR']** during the month of **June 2026**. The x-axis should represent the names of the processes, while the y-axis should display the rejection quantities. Individual bars should be displayed by location, with distinct colors or patterns used to represent each location, ensuring clarity and easy differentiation. Include a legend to identify the locations, and ensure that all names and segments are clearly visible. Annotate each bar with the exact rejection quantity, positioning the annotations appropriately for readability. Label the axes and include a descriptive title to maintain a professional and visually appealing presentation.

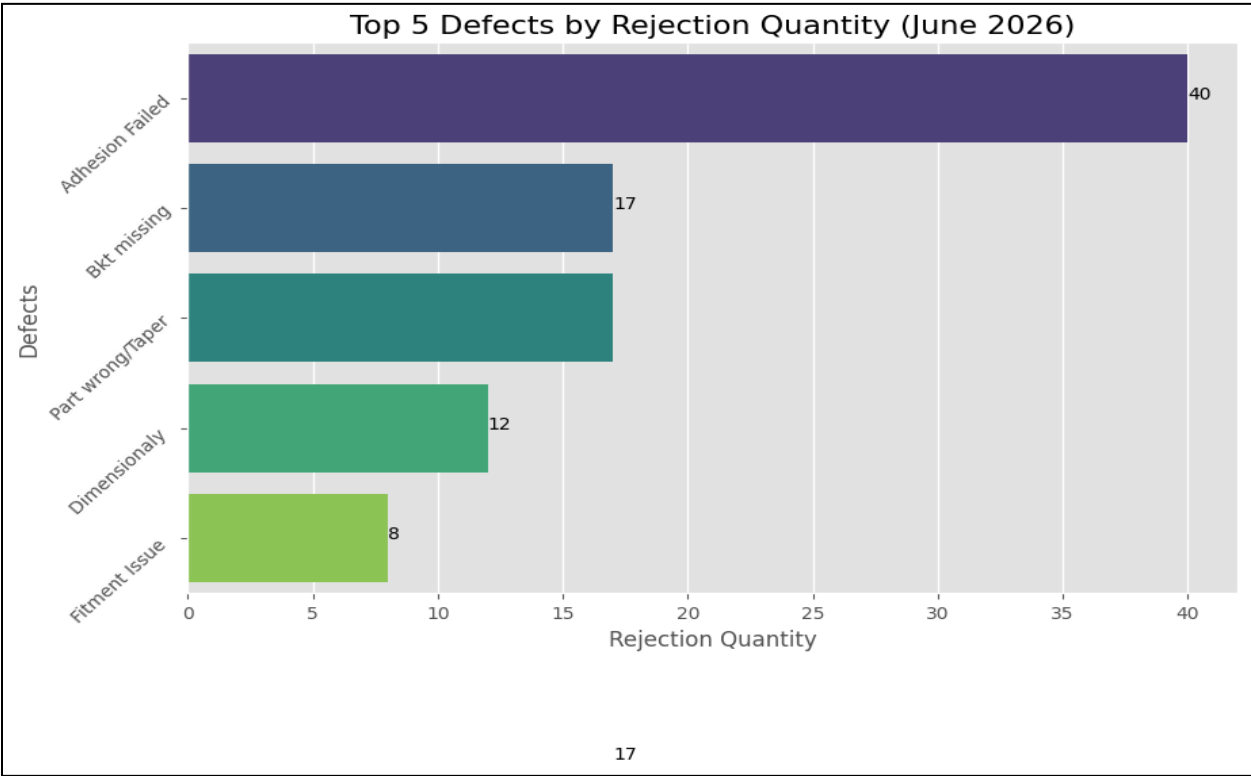


InsightEdge Observations:

- 1. **Dominant Process:** Assembly leads rejection quantity significantly at 44, far exceeding others.
- 2. **Minor Issues:** Machining has the lowest rejection count at 1, indicating minimal problems.
- 3. **Other Processes:** Press Shop and Welding show rejections of 12 and 5, reflecting moderate issues needing attention.

6. Most Impactful Root Causes (Overall)

Query: Create a professional horizontal bar graph to visualize the **top 5** defects for all locations **['PUN', 'LKN', 'JSR']** for the month of **June 2026**. The x-axis should represent the rejection quantities, while the y-axis should display the names of the defects. Each bar should correspond to a specific defect, with distinct colors or patterns ensuring clarity and easy differentiation. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability. Label the axes and ensure all the names are clearly visible, and include a descriptive title to create a professional and visually appealing presentation.

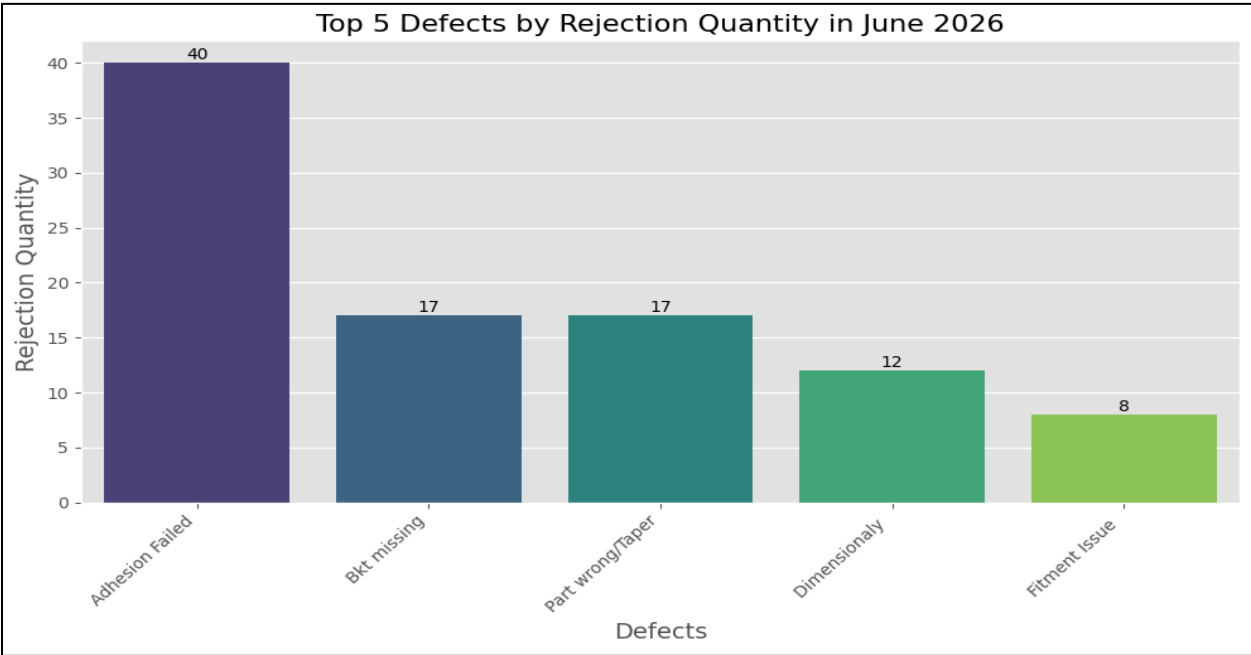


InsightEdge Observations:

- 1. **Main Defect:** Adhesion Failed significantly leads rejections with 40 instances, far surpassing others.
- 2. **Notable Comparison:** Bkt Missing and Part Wrong/Taper have close rejection numbers at 17 and 12, respectively.
- 3. **Minor Issues:** Dimensionality and Fitment Issues are the least concerning, with 12 and 8 rejections respectively.

7. Most Impactful Root Causes (Location-wise)

Query: Create a professional vertical column bar graph to visualize the **top 5** defects for each of the locations **['PUN', 'LKN', 'JSR']** for the month of **June 2026**. The y-axis should represent the rejection quantities, while the x-axis should display the names of the defects. Each bar should correspond to a specific defect, with distinct colors or patterns ensuring clarity and easy differentiation. Annotate each bar with the exact rejection quantity, ensuring the annotations are clear and positioned appropriately near the bars for readability. Label the axes and ensure all the names are clearly visible, and include a descriptive title to create a professional and visually appealing presentation.

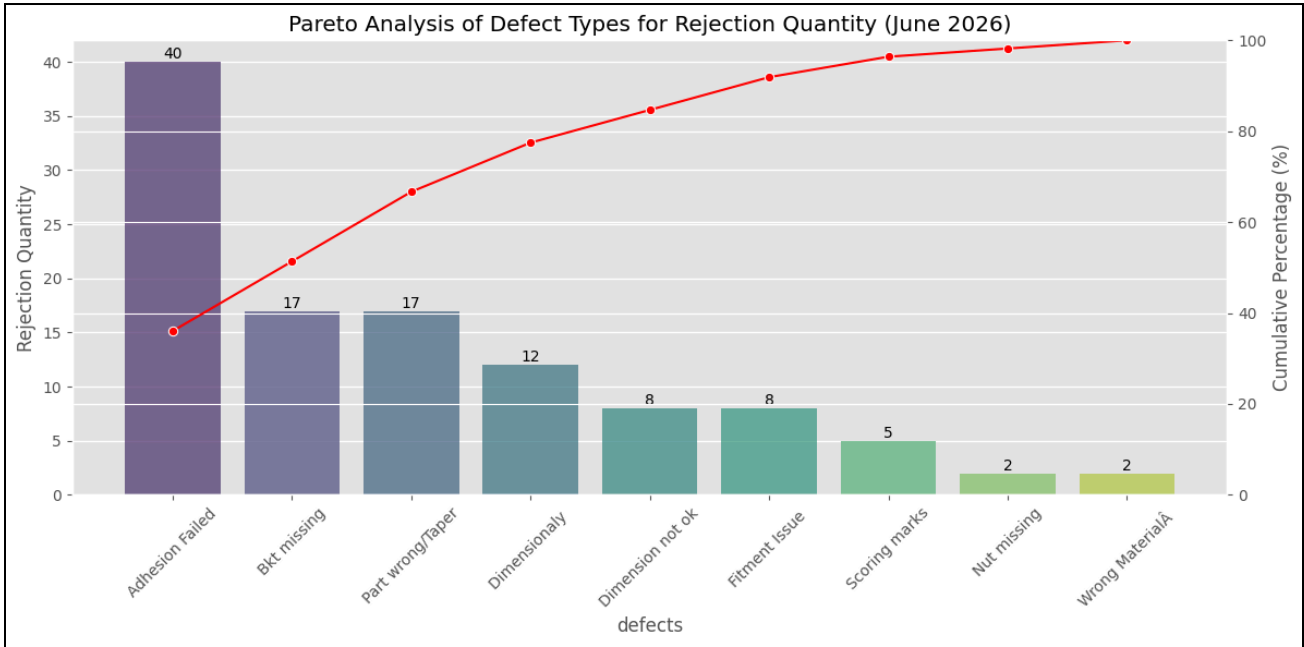


InsightEdge Observations:

- 1. **Dominant Defect:** "Adhesion Failed" accounts for the highest rejection quantity at 40, significantly leading the others.
- 2. **Equal Issues:** Both "Bkt missing" and "Part wrong/paper" show identical rejection counts of 17, indicating a need for focused improvement.
- 3. **Lower Rejections:** "Dimensionality" and "Fitment Issue" had lower rejection quantities of 12 and 8 respectively, suggesting less critical concerns.

8. Pareto analysis of Defects Types

Query: Create a Pareto analysis bar graph chart with a cumulative line of percentages for the top defect types in the customer complaints rejection quantity data for the latest month **June 2026**, where the x-axis displays the defect names vertically. Do not cut anything and make the visualization aesthetic.

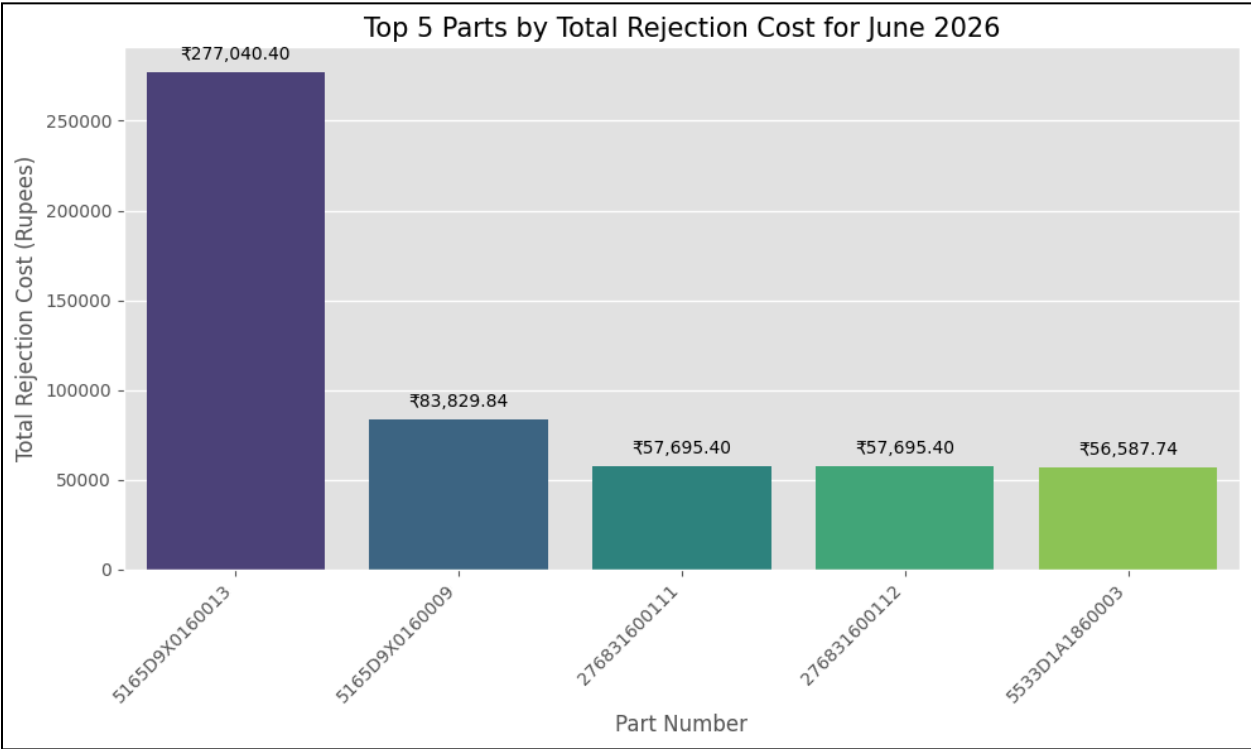


InsightEdge Observations:

- 1. **Major Defect:** Adhesion failed accounts for 36% of total rejections, highlighting a key quality issue.
- 2. **Cumulative Impact:** The first two defect types combined exceed 50% of rejections, indicating focus areas for improvement.
- 3. **Threshold Reached:** Cumulative rejections surpass 80% threshold by the sixth defect type, signaling immediate action needed.

9. Top Parts with Highest Rejection Cost

Query: Create a bar graph showing the **top 5** parts with total rejection cost across all locations **['PUN', 'LKN', 'JSR']** for **June 2026**. The x-axis must show part numbers and the y-axis the total rejection cost in rupees. Use vibrant colors and space x-axis labels well for visibility. Annotate each bar with rejection cost values.

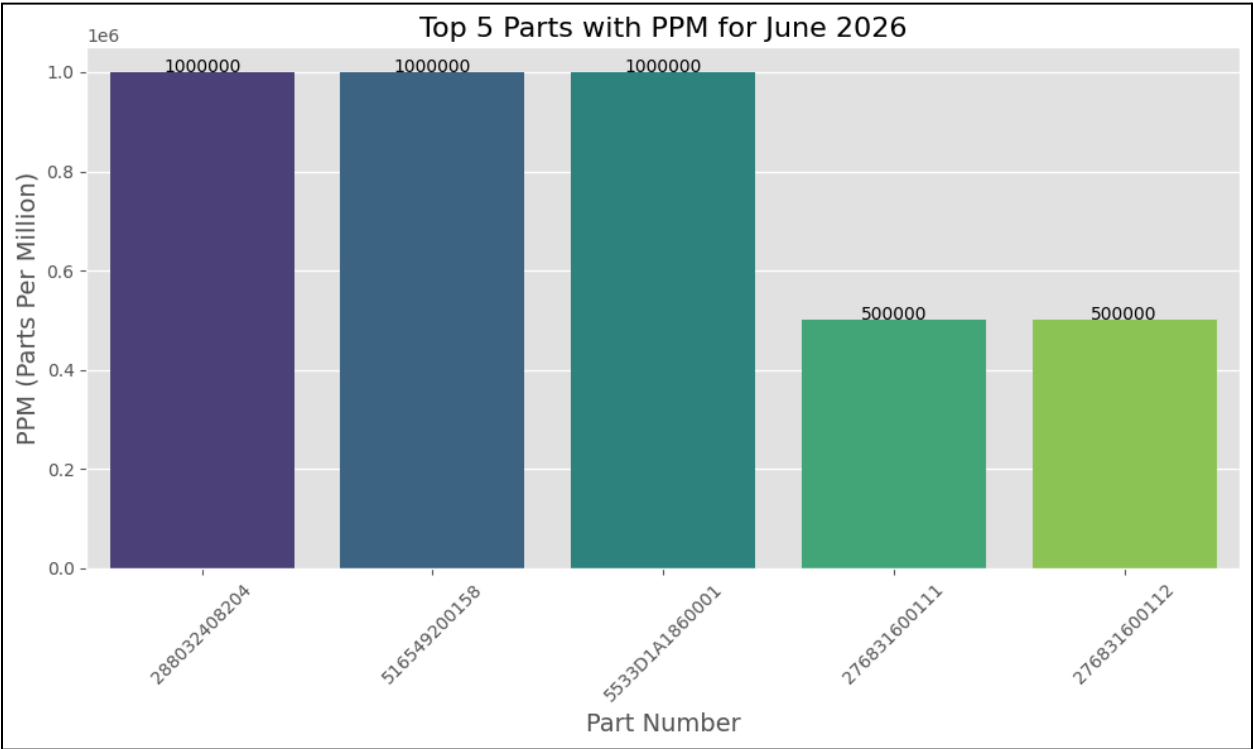


InsightEdge Observations:

- 1. **Dominant Cost:** Part 51659X016013 has a significantly higher rejection cost of ₹277,040.40 compared to others.
- 2. **Close Competition:** The rejection costs for parts 27683160011 and 27683160012 are tied at ₹57,695.40 each.
- 3. **Low Rejections:** The remaining parts show much lower total rejection costs, with the highest at ₹83,829.84 for part 51659X016009.

10. PPM

Query: Create a bar chart showing the **top 5** parts with PPM, calculated using $PPM = [(Rejection\ Quantity / Sale\ Quantity) \times 1,000,000]$, across all locations **['PUN', 'LKN', 'JSR']** for **June 2026**. The x-axis should represent the part numbers and the y-axis should represent the PPM value. Annotate the bars with PPM values using horizontal labels and use vibrant colors with professional visualization practices. If Sale Quantity is zero, treat PPM as 0 to avoid division by zero.



InsightEdge Observations:

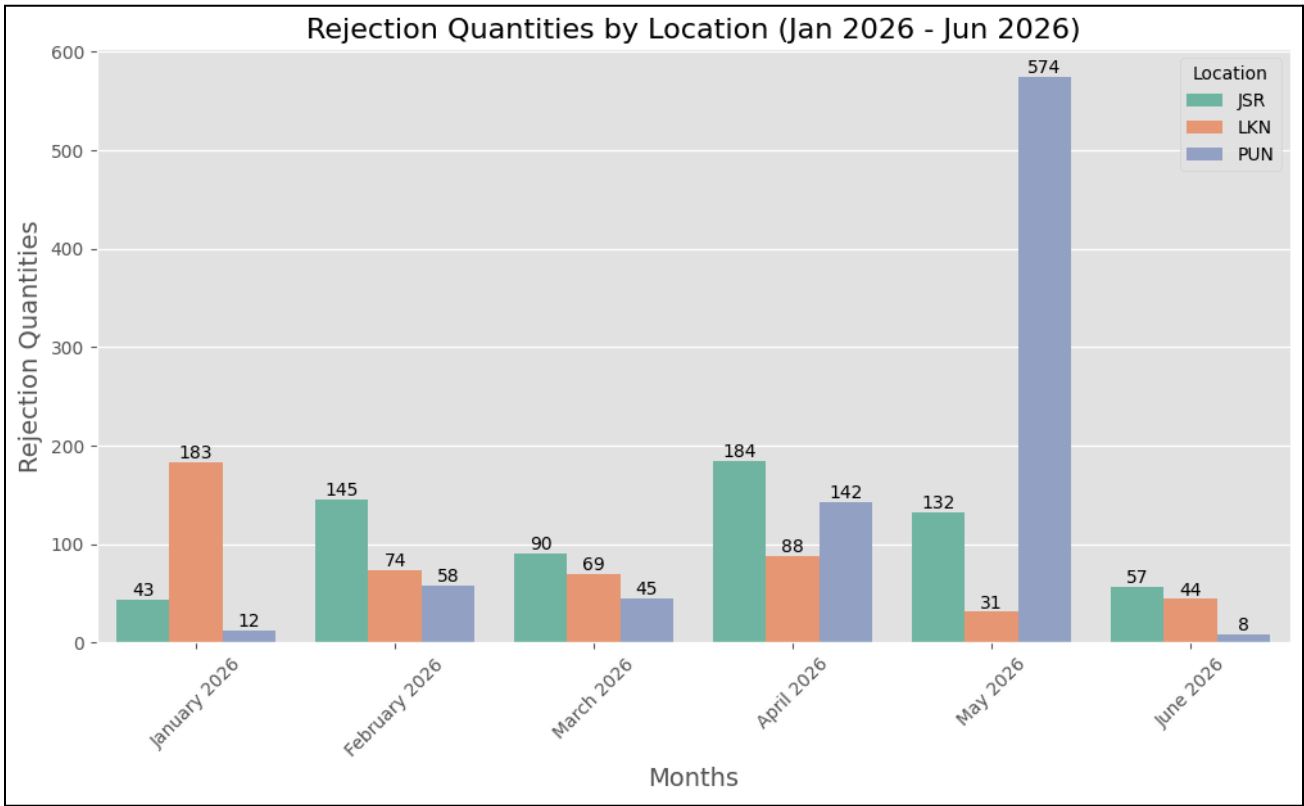
- 1. **PPM Leaderboard:** Three parts show identical PPM values of 1,000,000, indicating equal relevance.
- 2. **Significant Drop:** Two parts decline to 500,000 PPM, suggesting a disparity in performance.
- 3. **Uniformity Observed:** The top five parts highlight considerable variation in parts per million performance.

Agenda 4:

Trend analysis for Last 6 months
(including Current Month)

1. Location-Wise Summary

Query: Create a bar graph to represent the rejection quantities for three locations: ['PUN', 'LKN', 'JSR'], spanning the months from January 2026 to June 2026. Label the x-axis with the names of the months in the correct chronological order between January 2026 and June 2026, and the y-axis with the rejection quantities. Use distinct colors for each location to ensure clarity and differentiate the bars visually. Remove any black lines (error bars) and annotate each bar with its corresponding rejection quantity value, rounded to real numbers for enhanced readability and presentation.

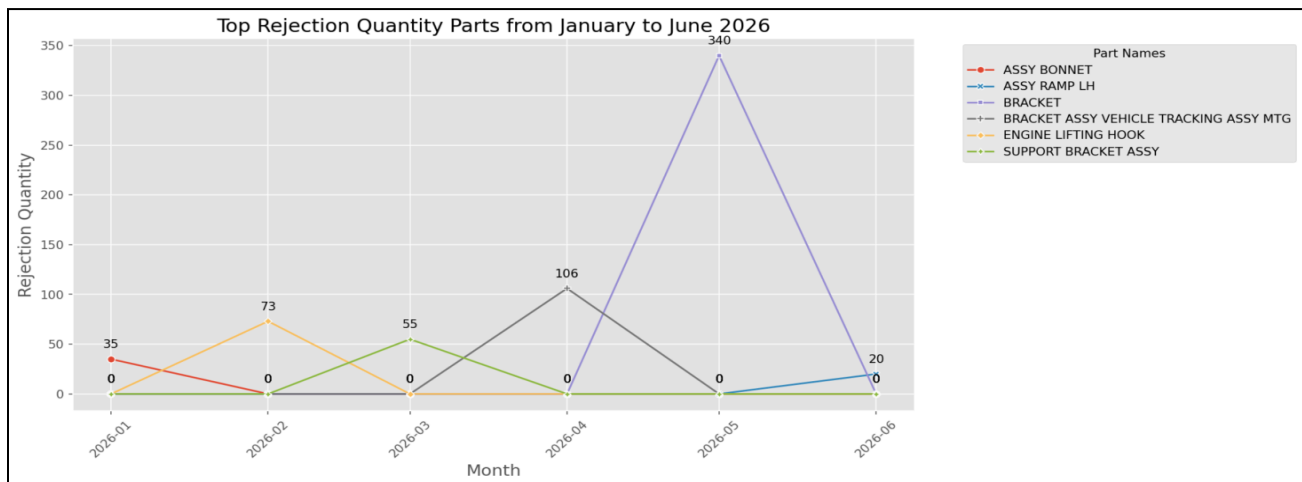


InsightEdge Observations:

- 1. **JSR Dominance:** JSR showed the highest rejection quantity in June with 574, significantly outpacing other locations.
- 2. **LKN Increase:** LKN rejections peaked at 184 in April, indicating a rising trend mid-year before declining.
- 3. **PUN Stability:** PUN exhibited consistent but lower rejection rates, with 44 in June, suggesting manageable issues.

2. Top Defective Parts

Query: Create a multi-line graph chart to track the part names with the highest rejection quantity for each month from **January 2026 to June 2026**. The graph should include parts, each representing the top rejection part for one month, and their rejection quantities should be tracked consistently across all months, regardless of their ranking in other months. The x-axis should show the months in strict chronological order between **January 2026** and **June 2026**, and the y-axis should represent the rejection quantities, explicitly showing values of 0 where applicable. Each part should be represented by a distinct line with appropriate markers and colors to ensure clarity. A legend should be included to identify the parts and should not cover the lines. Label the axes and title appropriately to maintain a professional and clean presentation. Annotate each points on line chart.

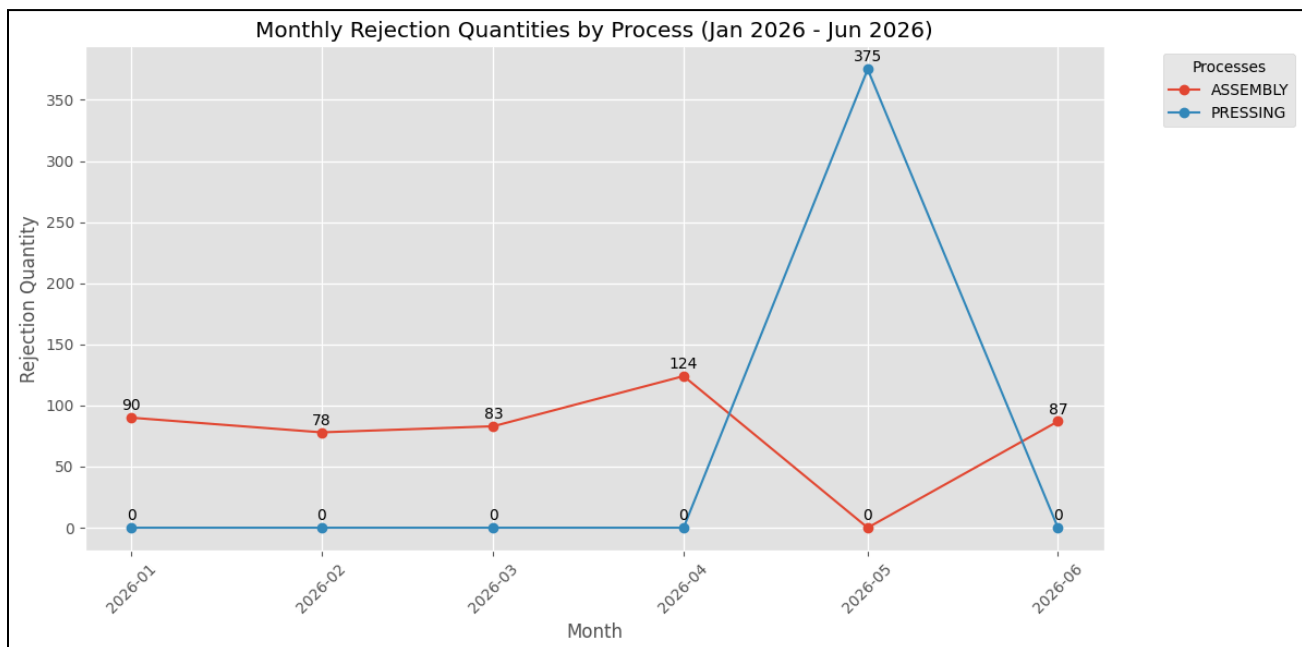


InsightEdge Observations:

- Major Rejection Spike:** BRAKE ASSY VEHICLE TRACKING ASSY MTG showed a sharp increase in rejection quantity, peaking at around **340 in May 2026**, indicating a major quality concern.
- Fluctuating Rejections:** ENGINE LIFTING HOOK recorded a significant spike of approximately 73 in February 2026, followed by a decline, showing inconsistent quality performance.
- Temporary Defect Rise:** SUPPORT BRACKET ASSY experienced a noticeable increase to around 55 in March 2026, suggesting a short-term production or assembly issue..
- Stable Performance in Other Parts:** Most other part names maintained relatively low rejection quantities throughout the timeline, reflecting more stable manufacturing quality.

3. Top Critical Processes

Query: Create a multi-line graph to track the processes with the highest rejection quantity for each month from **January 2026 to June 2026**. The graph should include processes, each representing the top rejection process for one of the months in the selected period, and these processes should be tracked consistently across all months, irrespective of their rankings in other months. The x-axis should represent the months, while the y-axis should display the rejection quantities. Each process should be represented by a distinct line with appropriate markers and colors, ensuring clarity and distinction. Include a legend to identify the processes, and label the axes and title appropriately to maintain a professional and clean visualization. Avoid using any additional highlights or emphasis on specific points. Annotate each points on line chart.

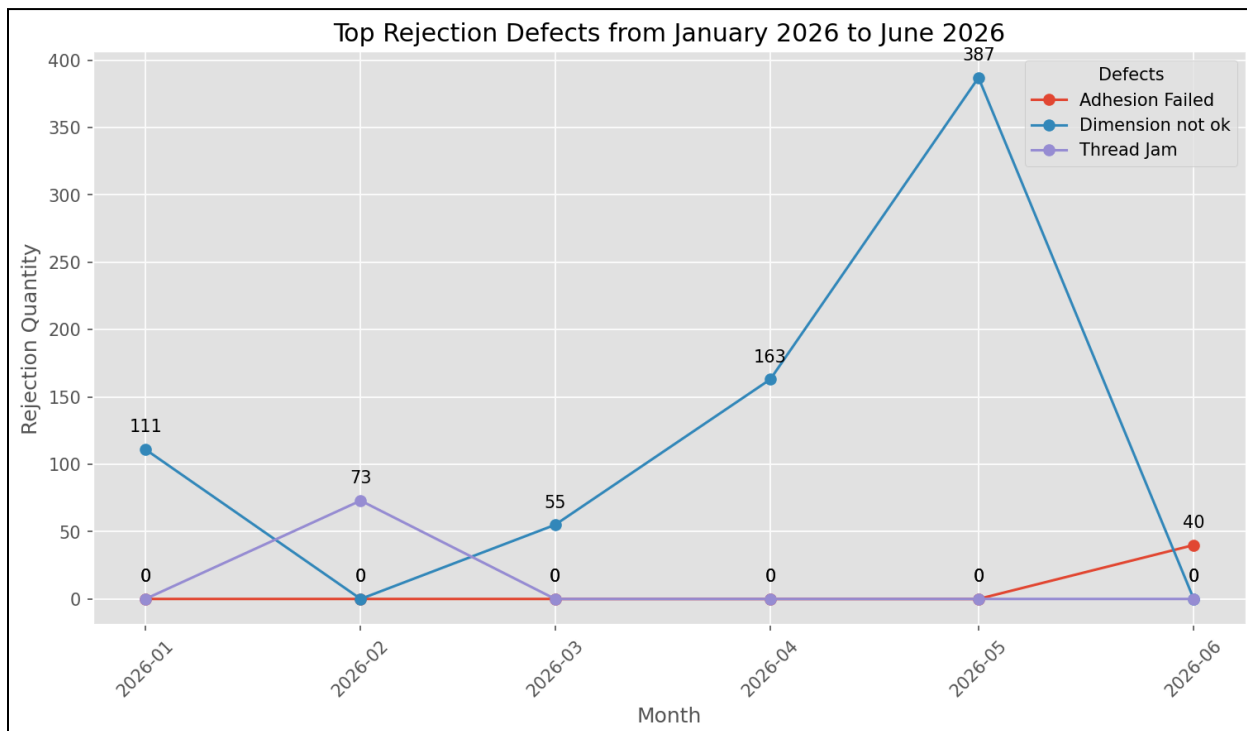


InsightEdge Observations:

- Overall Decrease:** The values show a significant decrease from the start to the end of the period analyzed.
- Consistent Low:** The lowest values remain consistent across the chart, indicating stable but low performance.
- Minor Fluctuations:** Slight fluctuations in the middle indicate temporary changes, suggesting factors influencing those periods.

4. Most Impactful Root Causes

Query: Create a multi-line graph chart to track the defects with the highest rejection quantity for each month from **January 2026 to June 2026**. The graph should include defects, each representing the top rejection defect for one month, and their rejection quantities should be tracked consistently across all months, regardless of their ranking in other months. The x-axis should show the months in strict chronological order between **January 2026** and **June 2026**, and the y-axis should represent the rejection quantities, explicitly showing values of 0 where applicable. Each defect should be represented by a distinct line with appropriate markers and colors to ensure clarity. A legend should be included to identify the top defects, and the axes and title should be visibly labeled to maintain a professional and clean presentation. Annotate each points on line chart.

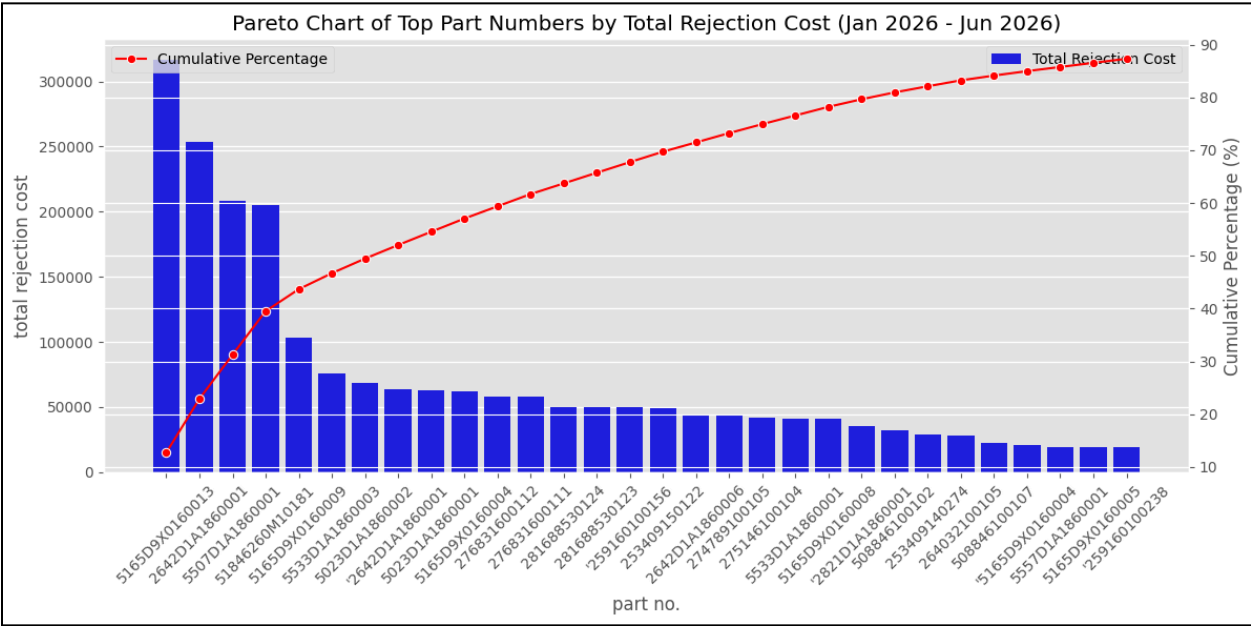


InsightEdge Observations:

- Adhesion Issue:** Adhesion failures peaked at 387 in May, indicating a critical defect trend.
- Overall Trend:** Rejection quantities consistently dropped to zero in April and May, suggesting a potential improvement.
- Monthly Variation:** Notable fluctuation in rejection rates, particularly from 111 in January to 163 in March, highlighting operational instability.

5. Pareto analysis for Total Rejection Cost

Query: Pareto chart showing top part numbers by total rejection cost with cumulative percentage line for the period from **January 2026 to June 2026** across locations **['PUN', 'LKN', 'JSR']**. Include labels only for the **top 30** parts. Do not annotate points with rejection quantity or percentage. Tilt x-axis labels by 45 degrees.

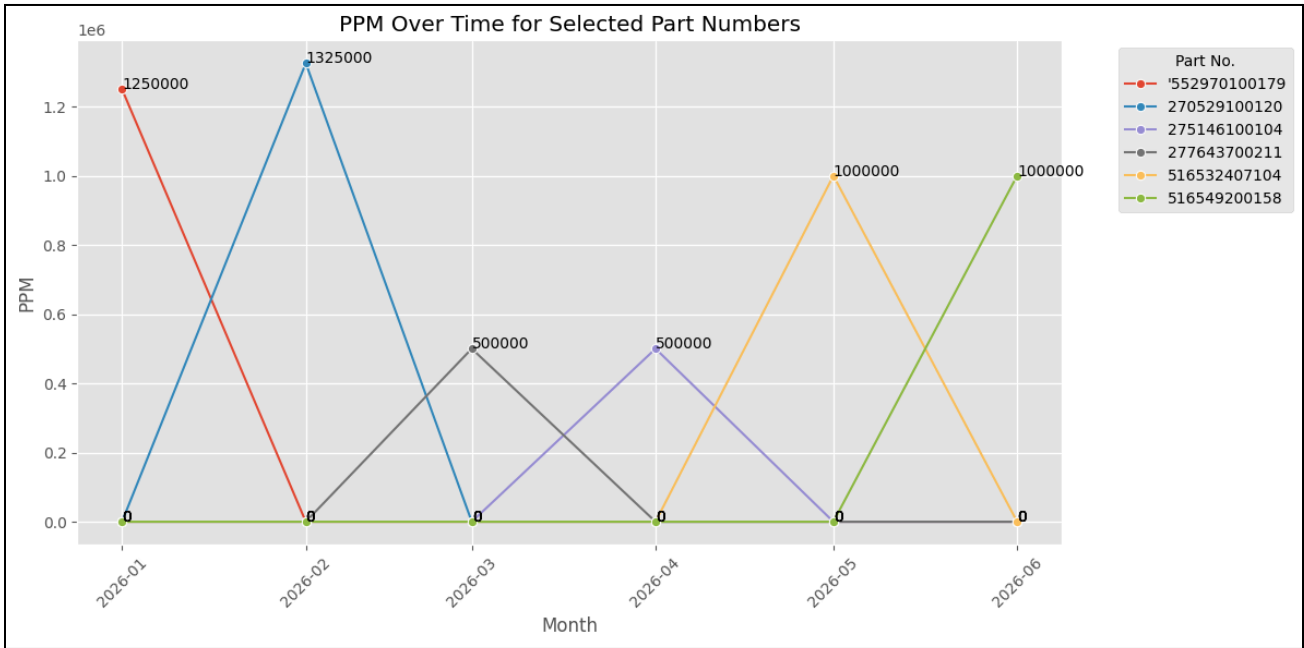


InsightEdge Observations:

- 1. **High Rejection Costs:** The top few part numbers account for a significant share of total rejection costs, indicating a targeted area for improvement.
- 2. **Cumulative Increase:** The cumulative percentage curve shows that a small number of parts contribute to most of the rejection costs, aligning with the Pareto principle.
- 3. **Declining Contributions:** The rejection costs for additional parts decrease sharply after the top entries, suggesting that focusing on these parts could yield the highest impact.

6. PPM Trends

Query: Follow the instructions step by step: For every month from **January 2026 to June 2026**, select the part number with the highest PPM, selecting a single part number for each month. PPM = (Rejection Quantity × 1,000,000) / Sale Quantity . For every selected part number, calculate its PPM across all months in the selected period. If no data is found for a month, assign a value of 0. Structure the results as a table where each row represents a part number and each column represents a month, with values representing PPM. Generate a multi-line chart using this data. The x-axis represents months (convert to datetime before plotting), and the y-axis represents PPM. Each line represents a part number.



InsightEdge Observations:

- 1. **Major PPM Spikes:** Part No. **270529100120** showed extremely high PPM spikes in **December 2025** and **February 2026**, reaching above **1.3 million**, indicating severe quality issues requiring immediate investigation.
- 2. **Sharp Monthly Variations:** Several part numbers displayed sudden rises and drops in PPM values across months, reflecting inconsistent production or process stability.
- 3. **Stable Low PPM Periods:** Most part numbers maintained relatively low PPM values during several months, suggesting stable quality performance outside peak periods.